



PAN3029/PAN 3060 Series

Product Manual

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Shanghai Panchip Microelectronics Co., Ltd.

PAN3029/PAN 3060 Series

PAN3029 / PAN3060 is a low-power long-distance wireless transceiver chip using ChirpIoT™ modulation and demodulation technology. It supports half-duplex wireless communication and operates in the frequency bands of 137~187MHz, 205~281MHz, 408~550MHz and 816~1110MHz. The chip has the characteristics of high anti-interference, high sensitivity, low power consumption and ultra-long transmission distance. It has a maximum sensitivity of -143dBm and a maximum output power of 20dBm, which produces an industry-leading link budget, making it the best choice for long-distance transmission and applications with extremely high reliability requirements.

Key Features

- Operating frequency bands:
137~187MHz, 205~281MHz, 408~550MHz,
816~1110MHz
- Modulation mode: ChirpIoT™
- Transmit output power: -30dBm~20dBm
- Maximum link budget: 163dB
- Sensitivity as low as: -143dBm@62.5kHz
- Operating current
 - Deep sleep current: 260nA
 - Receive current: 4.1mA@DCDC mode
 - Transmit current:
 - 96mA@20dBm
 - 13mA@-2dBm
- Supported bandwidth
 - PAN3029: 62.5kHz、125kHz、250kHz、500kHz
 - PAN3060: 125kHz、250kHz、500kHz
- SF factor support, automatic SF identification
 - PAN3029: 5~12
 - PAN3060: 5~9
- Support code rate: 4/5, 4/6, 4/7, 4/8
- Support CAD function
- Support low rate mode: 0.08~59.9kbps

- Support 4-wire SPI / 3-wire SPI / I2C configuration interface, support 4 GPIOs
- Operating voltage:
1.8V ~3.6V (DCDC mode 2V~3.6V)
- Operating temperature: -40°C~85°C
- Package: QFN28 (4×4mm)

Typical Applications

- Smart factory
- Smart water
- Smart agriculture
- Smart healthcare
- Smart community
- Smart fire protection

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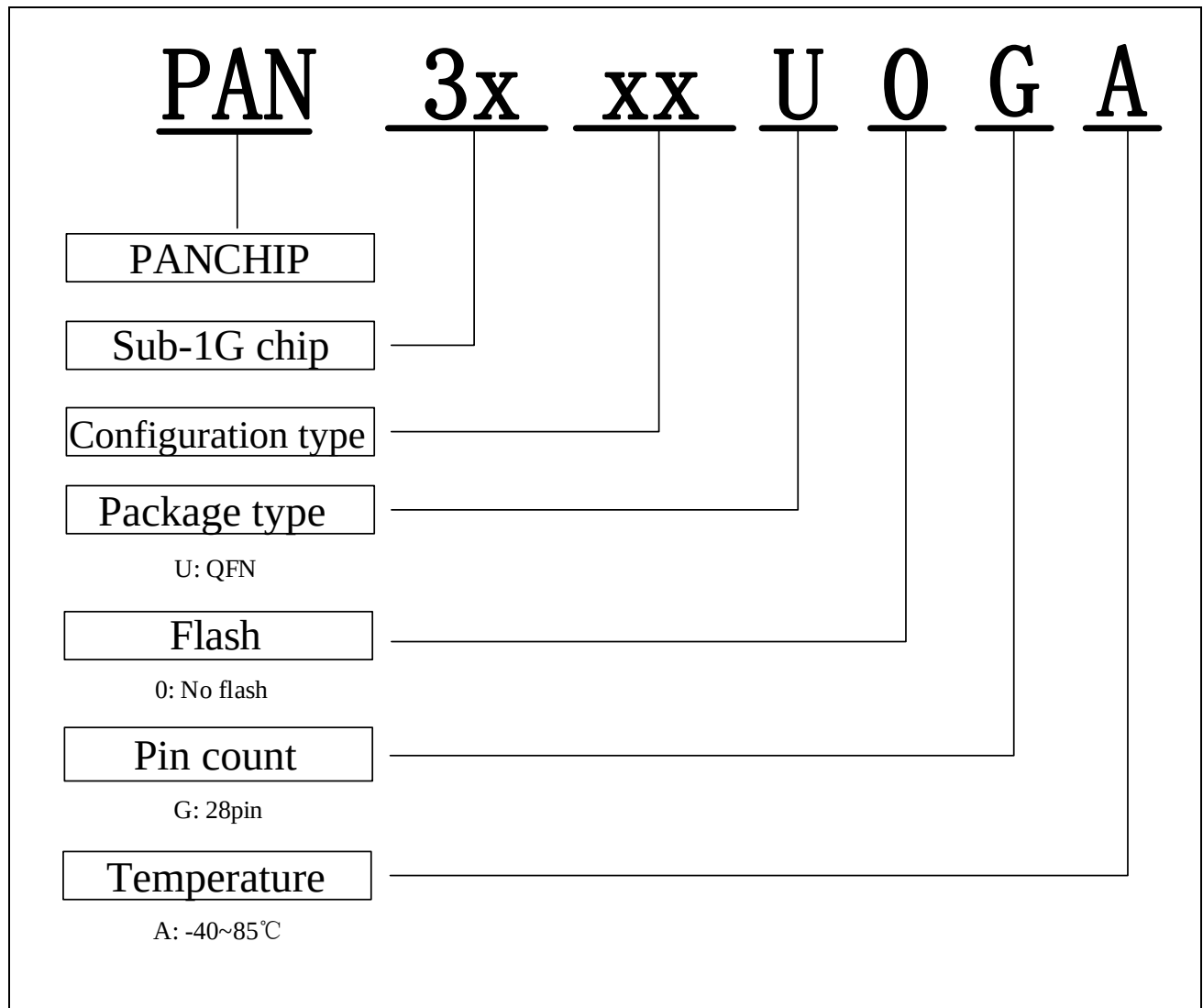
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1 Naming convention



2 Ordering Information

Product Model	Chip Type	Package	Pin count	IO	Bandwidth	SF	Bit Rate	Temperature	Packaging
PAN3029 U0GA	Sub-1G	QFN	28	4	62.5kHz 125kHz 250kHz 500kHz	SF5~SF12	0.08~ 59.9kbps	-40~85°C	Tape & Reel
PAN3060 U0GA	Sub-1G	QFN	28	4	125kHz 250kHz 500kHz	SF5~SF9	0.5~ 59.9kbps	-40~85°C	Tape & Reel

Before ordering, please consult sales for the latest mass production information.

This manual describes the models based on the largest package and most complete functions. For configuration differences between models, please refer to the ordering information on this page.

3 System structure block diagram

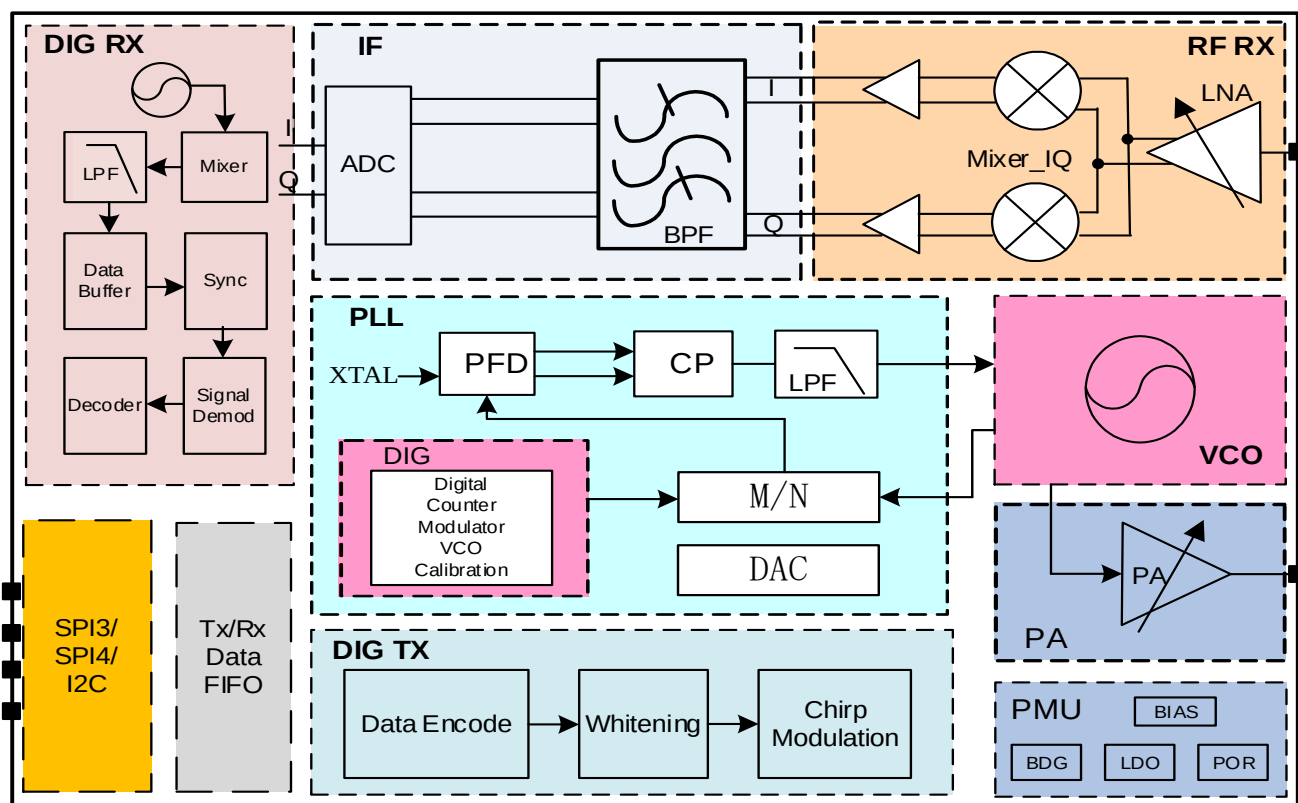


Figure 3-1 System Block Diagram

4 Pin Definition and Description

4.1 Pin Diagram

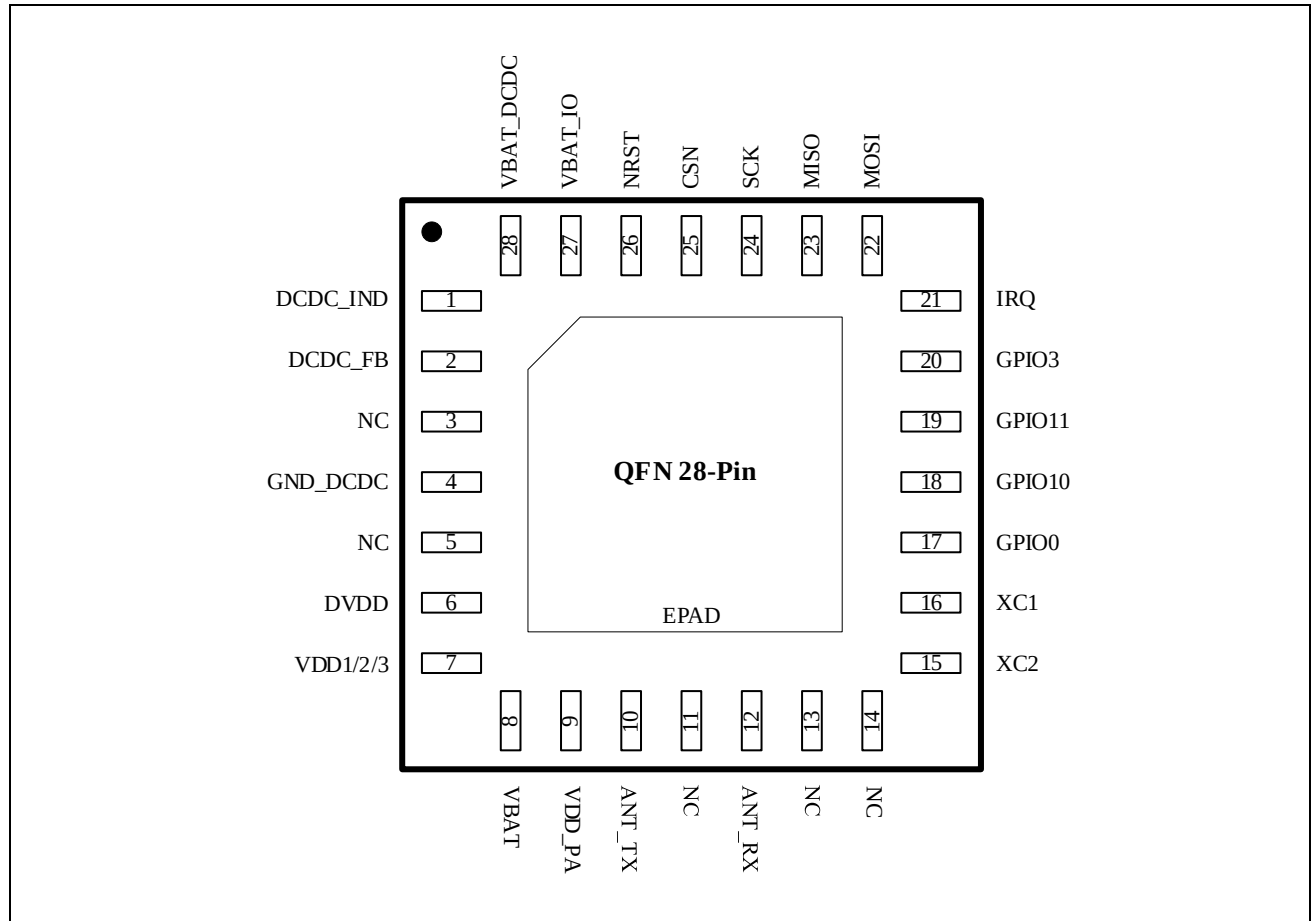


Figure 4-1 Pin Assignment

4.2 Pin Description

Table 4-1 Pin Description

No.	Symbol	Type	Function
1	DCDC_IND	AO	Internal DCDC output, connects to an external series inductor
2	DCDC_FB	AI	Internal DCDC feedback input, connects to VDD1/2/3 in DCDC mode
3	NC		For optimal performance, connect to GND_DCDC
4	GND_DCDC	G	Analog ground
5	NC		For optimal performance, connect to GND_DCDC
6	DVDD	P	Digital power, LDO output
7	VDD1/2/3	P	Analog power, connects to VFB in DCDC mode, connects to the main power supply in non-DCDC mode
8	VBAT	P	Analog power, connects to the main power supply
9	VDD_PA	P	PA power LDO output
10	ANT_TX	AO	Transmitter PA output
11	NC		For optimal performance, connect to GND
12	ANT_RX	AI	Receiver LNA input
13	NC		For optimal performance, connect to GND
14	NC		For optimal performance, connect to GND
15	XC2	AI	Crystal Oscillator Input
16	XC1	AO	Crystal Oscillator Output
17	GPIO0	I/O	Digital IO
18	GPIO10	I/O	Digital IO
		O	External PA control signal
19	GPIO11	I/O	Digital IO
		O	Channel status indicator signal
20	GPIO3	I/O	Digital IO
21	IRQ	O	Interrupt signal
22	MOSI	I	SPI data input signal
23	MISO	O	SPI data output signal
24	SCK	I	SPI serial clock
25	CSN	I	SPI chip select signal
26	NRST	I	Reset
27	VBAT_IO	P	Digital GPIO power supply, connected to the main power supply
28	VBAT_DCDC	P	DCDC power supply, connected to the main power supply



PAN3029 / PAN3060 series wireless transceiver chips

29	E-PAD	G	GND pad on the bottom of the chip, needs to be grounded
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5 Electrical characteristics

All parameters in this section are based on testing at $T_A=25^{\circ}\text{C}$ and a 3.3V supply voltage, unless otherwise specified.

5.1 Absolute Maximum Ratings

Table 5-1 Absolute Maximum Ratings

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
VDD	VDD1/VDD2/VDD3/VBAT/VBAT_IO	Frequency: 490MHz	-0.3	3.3	3.6	V
V _I	Input Voltage		-0.3	-	VDD	V
V _O	Output Voltage		VSS	-	VDD	V
T _{OP}	Operating Temperature		-40	-	85	°C
T _{STG}	Storage Temperature		-55	-	125	°C

Caution:

Exceeding one or more of the limiting values may cause permanent damage to the chip.

This device is electrostatically sensitive. Please comply with protective regulations when handling.

5.2 DC characteristics

Test conditions:

Frequency: 490MHz

Table 5-2 Voltage and current (frequency 490MHz)

Symbol	Parameter	Min	Typ	Max	Unit	Test Conditions
VDD	Power Supply	1.8	3.3	3.6	V	$T_A=25^{\circ}\text{C}$, LDO mode
		2	3.3	3.6	V	$T_A=25^{\circ}\text{C}$, DCDC mode
VSS	Ground	-	0	-	V	-
I _{DeepSleep}	Deep Sleep Current	-	260	-	nA	-
I _{Sleep}	Sleep Current	-	620	-	nA	-
I _{TX,20dBm}	TX Mode Operating Current	-	96	-	mA	20dBm Output Power
I _{TX,18dBm}	TX Mode Operating Current	-	77	-	mA	18dBm Output Power
I _{TX,-2dBm}	TX Mode Operating Current	-	13	-	mA	-2dBm Output Power
I _{RX,LDO}	RX Mode Operating Current	-	7.8	-	mA	LDO Mode, maximum LNA gain
I _{RX,DCDC}	RX Mode Operating Current	-	4.1	-	mA	DCDC mode, maximum LNA gain
V _{OH}	Output High Voltage	VDD-0.3	-	VDD	V	-
V _{OL}	Output Low Voltage	VSS	-	VSS+0.3	V	-

V _{IH}	Input high level voltage	0.8*VDD	-	-	V	-
V _{IL}	Input low level voltage	-	-	0.2*VDD	V	-
SPI_rate	SPI rate	-	-	10	Mbps	-

Test conditions:

Frequency: 915MHz

Table 5-3 Voltage and current (frequency 915MHz)

Symbol	Parameter	Min	Typ	Max	Unit	Test Conditions
VDD	Power Supply	1.8	3.3	3.6	V	T _A = 25°C, LDO mode
		2	3.3	3.6	V	T _A = 25°C, DCDC mode
VSS	Ground	-	0	-	V	-
I _{DeepSleep}	Deep Sleep Current	-	260	-	nA	-
I _{Sleep}	Sleep Current	-	620	-	nA	-
I _{TX,20dBm}	TX Mode Operating Current	-	100	-	mA	20dBm Output Power
I _{TX,18dBm}	TX Mode Operating Current	-	81	-	mA	18dBm Output Power
I _{TX,-2dBm}	TX Mode Operating Current	-	15	-	mA	-2dBm Output Power
I _{RX,LDO}	RX Mode Operating Current	-	7.8	-	mA	LDO Mode, maximum LNA gain
I _{RX,DCDC}	RX Mode Operating Current	-	4.1	-	mA	DCDC mode, maximum LNA gain
V _{OH}	Output High Voltage	VDD-0.3	-	VDD	V	-
V _{OL}	Output Low Voltage	VSS	-	VSS+0.3	V	-
V _{IH}	Input high level voltage	0.8*VDD	-	-	V	-
V _{IL}	Input low level voltage	-	-	0.2*VDD	V	-
SPI_rate	SPI rate	-	-	10	Mbps	-

5.3 RF Characteristics

Test conditions:

Frequency: 490MHz

Error Correction Code = 4/8

Packet Error Rate = 0.001

Payload Length = 10 Bytes

Table 5-4 General RF Characteristics (Frequency 490MHz)

Symbol	Parameter	Condition	Min	Typ	Max	Unit
Common characteristics						
F _{op}	Operating Frequency	-	137	-	187	MHz
		-	205	-	281	MHz
		-	408	-	550	MHz
		-	816	-	1110	MHz
F _{xtal}	Crystal Oscillator Frequency	-	-	32	-	MHz
R _S	Crystal Series Resistance	-	-	30	50	
C _{FOOT}	Crystal External Capacitance	-	8	15	22	pF
C _{LOAD}	Crystal Load Capacitance	-	6	10	12	pF
F _{TOL}	Initial Frequency Tolerance	-	-	±10	-	ppm
BR	Bit Rate (PAN3029 Series)	-	0.08	-	59.9	kbps
	Bit Rate (PAN3060 Series)	-	0.5	-	59.9	kbps
Transmitter						
P _{LPWAN}	Output Power	-	-30	-	20	dBm
Receiver						
RF_62.5 (LDO)	RF sensitivity, long range mode, maximum LNA gain, using separate RX/TX channels 62.5 kHz bandwidth	SF = 7	-	-129	-	dBm
		SF = 10	-	-137	-	
		SF = 12	-	-143	-	
RF_125 (LDO)	RF sensitivity, long range mode, maximum LNA gain, using separate RX/TX channels 125 kHz bandwidth	SF = 7	-	-126	-	dBm
		SF = 10	-	-134	-	
		SF = 12	-	-140	-	
RF_250 (LDO)	RF sensitivity, long range mode, maximum LNA gain, using separate RX/TX channels 250 kHz bandwidth	SF = 7	-	-124	-	dBm
		SF = 10	-	-132	-	
		SF = 12	-	-137	-	
RF_500 (LDO)	RF sensitivity, long range mode, maximum LNA gain, using separate RX/TX channels 500 kHz bandwidth	SF = 7	-	-121	-	dBm
		SF = 10	-	-129	-	
		SF = 12	-	-134	-	
RF_62.5 (DCDC)	RF sensitivity, long range mode, maximum LNA gain, using separate RX/TX channels 62.5kHz bandwidth	SF = 7	-	-125	-	dBm
		SF = 10	-	-133	-	
		SF = 12	-	-139	-	
RF_125 (DCDC)	RF sensitivity, long range mode, maximum LNA gain, using separate RX/TX channels 125 kHz bandwidth	SF = 7	-	-122	-	dBm
		SF = 10	-	-130	-	
		SF = 12	-	-137	-	
RF_250	RF sensitivity, long range mode, maximum	SF = 7	-	-120	-	dBm

(DCDC)	LNA gain, using separate RX/TX channels 250 kHz bandwidth	SF = 10 SF = 12	- -	-128 -134	- -	
RF_500 (DCDC)	RF sensitivity, long range mode, maximum LNA gain, using separate RX/TX channels 500 kHz bandwidth	SF = 7 SF = 10 SF = 12	- - -	-117 -125 -130	- - -	dBm

Test conditions:

Frequency: 915MHz

Error Correction Code = 4/8

Packet Error Rate = 0.001

Payload Length = 10 Bytes

Table 5-5 General RF Characteristics (Frequency 915MHz)

Symbol	Parameter	Condition	Min	Typ	Max	Unit
Common characteristics						
F _{op}	Operating Frequency	-	137	-	187	MHz
		-	205	-	281	MHz
		-	408	-	550	MHz
		-	816	-	1110	MHz
F _{xtal}	Crystal Oscillator Frequency	-	-	32	-	MHz
R _s	Crystal Series Resistance	-	-	30	50	
C _{FOOT}	Crystal External Capacitance	-	8	15	22	pF
C _{LOAD}	Crystal Load Capacitance	-	6	10	12	pF
F _{TOL}	Initial Frequency Tolerance	-	-	±10	-	ppm
BR	Bit Rate (PAN3029 Series)	-	0.08	-	59.9	kbps
	Bit Rate (PAN3060 Series)	-	0.5	-	59.9	kbps
Transmitter						
P _{LPWAN}	Output Power	-	-30	-	20	dBm
Receiver						
RF_62.5 (LDO)	RF sensitivity, long range mode, maximum LNA gain, using separate RX/TX channels 62.5 kHz bandwidth	SF = 7	-	-128	-	dBm
		SF = 10	-	-136	-	
		SF = 12	-	-141	-	
RF_125 (LDO)	RF sensitivity, long range mode, maximum LNA gain, using separate RX/TX channels 125 kHz bandwidth	SF = 7	-	-125	-	dBm
		SF = 10	-	-132	-	
		SF = 12	-	-138	-	
RF_250 (LDO)	RF sensitivity, long range mode, maximum LNA gain, using separate RX/TX channels 250 kHz bandwidth	SF = 7	-	-121	-	dBm
		SF = 10	-	-129	-	
		SF = 12	-	-135	-	
RF_500 (LDO)	RF sensitivity, long range mode, maximum LNA gain, using separate RX/TX channels 500 kHz bandwidth	SF = 7	-	-119	-	dBm
		SF = 10	-	-127	-	
		SF = 12	-	-132	-	
RF_62.5 (DCDC)	RF sensitivity, long range mode, maximum LNA gain, using separate RX/TX channels 62.5kHz bandwidth	SF = 7	-	-128	-	dBm
		SF = 10	-	-136	-	
		SF = 12	-	-140	-	

RF_125 (DCDC)	RF sensitivity, long range mode, maximum LNA gain, using separate RX/TX channels 125 kHz bandwidth	SF = 7	-	-125	-	dBm
		SF = 10	-	-132	-	
		SF = 12	-	-137	-	
RF_250 (DCDC)	RF sensitivity, long range mode, maximum LNA gain, using separate RX/TX channels 250 kHz bandwidth	SF = 7	-	-120	-	dBm
		SF = 10	-	-129	-	
		SF = 12	-	-135	-	
RF_500 (DCDC)	RF sensitivity, long range mode, maximum LNA gain, using separate RX/TX channels 500 kHz bandwidth	SF = 7	-	-118	-	dBm
		SF = 10	-	-126	-	
		SF = 12	-	-130	-	

6 Frequency synthesizer

The chip integrates high-performance RF transceiver circuits and supports multiple Sub-1GHz frequency bands such as 150M/230M/400M/800M.

The chip contains a high-performance, low phase noise, fully integrated fractional frequency synthesizer. The frequency synthesizer is used to generate the LO frequency used by the down-conversion mixer in the receive mode. In the transmit mode, it is also used to directly generate the modulated RF carrier. The fractional frequency synthesizer provides excellent phase noise performance, frequency resolution less than 100Hz, and has low power consumption. The frequency synthesizer has a fast frequency stabilization function, which can complete the receive and transmit frequency stabilization in a very short time to optimize the chip power consumption.

7 Modem

The chip utilizes a modem that supports ChirpIoT™ transmission. This achieves an ultra-high communication link budget through superior spread-spectrum modulation and forward error correction technologies. Compared to traditional FSK-based modulation schemes, this increases the range and stability of the radio communication link.

The ChirpIoT™ modem significantly enhances interference immunity. Compared to FSK modulation, the ChirpIoT™ modem suppresses co-channel interference by up to 19dB, meaning that the interference signal energy can be 19dB higher than the ChirpIoT™ signal. This significantly ensures that the ChirpIoT™ signal can coexist easily with other modulation systems and be widely used in hybrid network communications.

7.1 Modulation parameters

The chip optimizes the transmission performance of the ChirpIoT™ signal by configuring different modulation parameters, achieving a balance between transmission rate, transmission distance, and transmission reliability. Specific parameters are as follows:

- Signal Bandwidth (BW)
- Spreading Factor (SF)
- Coding Rate (CR)
- Low Data Rate (LDR)

7.1.1 Signal Bandwidth

The PAN3029 supports four signal bandwidths: 500kHz, 250kHz, 125kHz, and 62.5kHz.

The PAN3060 supports three signal bandwidths: 500kHz, 250kHz, and 125kHz.

Different signal bandwidths can be selected through register configuration.

7.1.2 Spreading Factor

The PAN3029 supports eight different spreading factors, from SF5 to SF12, while the PAN3060 supports five different spreading factors, from SF5 to SF9. A larger spreading factor reduces the data payload rate.

7.1.3 Coding Rate

The chip uses forward error correction technology and supports four different coding rates: CR45, CR46, CR47, and CR48. This forward error correction technology corrects bit errors within a certain range, improving transmission reliability.

7.1.4 Low Rate Mode

Low rate mode is a communication mode of ChirpIoT™ modulation technology. It improves signal transmission reliability by reducing the number of valid bits of information carried per unit time.

7.2 Frame Structure

The frame structure of the ChirpIoT transmission signal consists of four parts: signal Preamble, Header, Payload, and CRC, as shown in the figure below.



Figure 7-1 PHY Frame Structure

8 MAC Design

8.1 Transmitter and receiver modes

The chip's transmitter has two unique operating modes, and the receiver has three unique operating modes.

Transmitter Operating Modes:

- Single-Packet Transmit Mode

- Continuous Transmit Mode

Receiver Operating Modes:

- Single-Packet Receive Mode

- Single-Packet Receive Mode with Timeout

- Continuous Receive Mode

8.2 Transmitter Mode

The MAC transmit mode has two modes: single-packet transmit and continuous transmit.

These modes are selected using the mode select register. After selecting TX mode in the mode select register, the state machine enters the TX ready state, waits for the FIFO to be filled with data, and then enters the TX transmit state.

When the MAC enters TX transmit mode, the power control module successively turns on analog circuits such as the LDO module, PLL module, and PA. It then sends a TX start signal to the modem module, initiating data transmission.

After data transmission is complete, the LDO module, PLL module, PA, and other analog circuits are sequentially turned off. Then send an IRQ signal to the MCU. After the MCU clears the IRQ, you can configure the REG_OPT_MODE register to enter the STB3 mode, ending a TX process.

8.2.1 Single packet transmit mode

Figure 8-1 shows the state flow diagram for single-packet transmission mode.

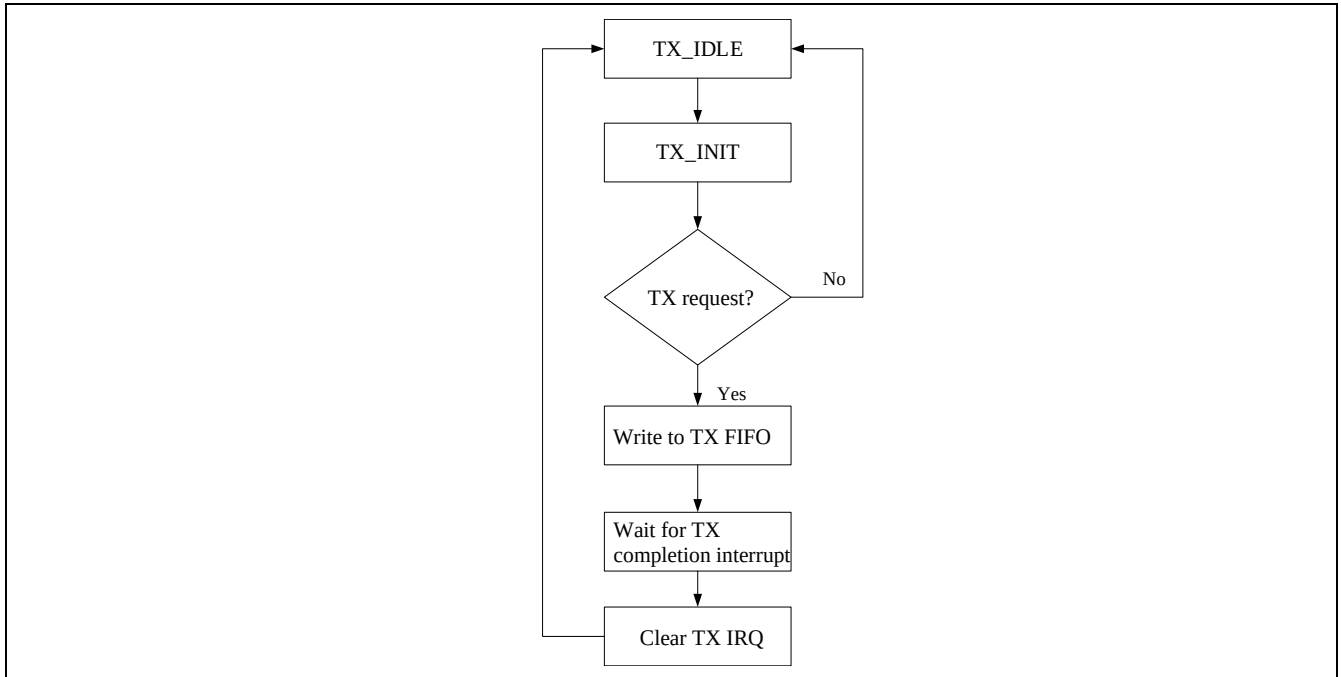


Figure 8-1 Single-Packet Transmit Mode State Flowchart

8.2.2 Continuous transmit mode

Figure 8-2 shows the state flow diagram for continuous transmit mode.

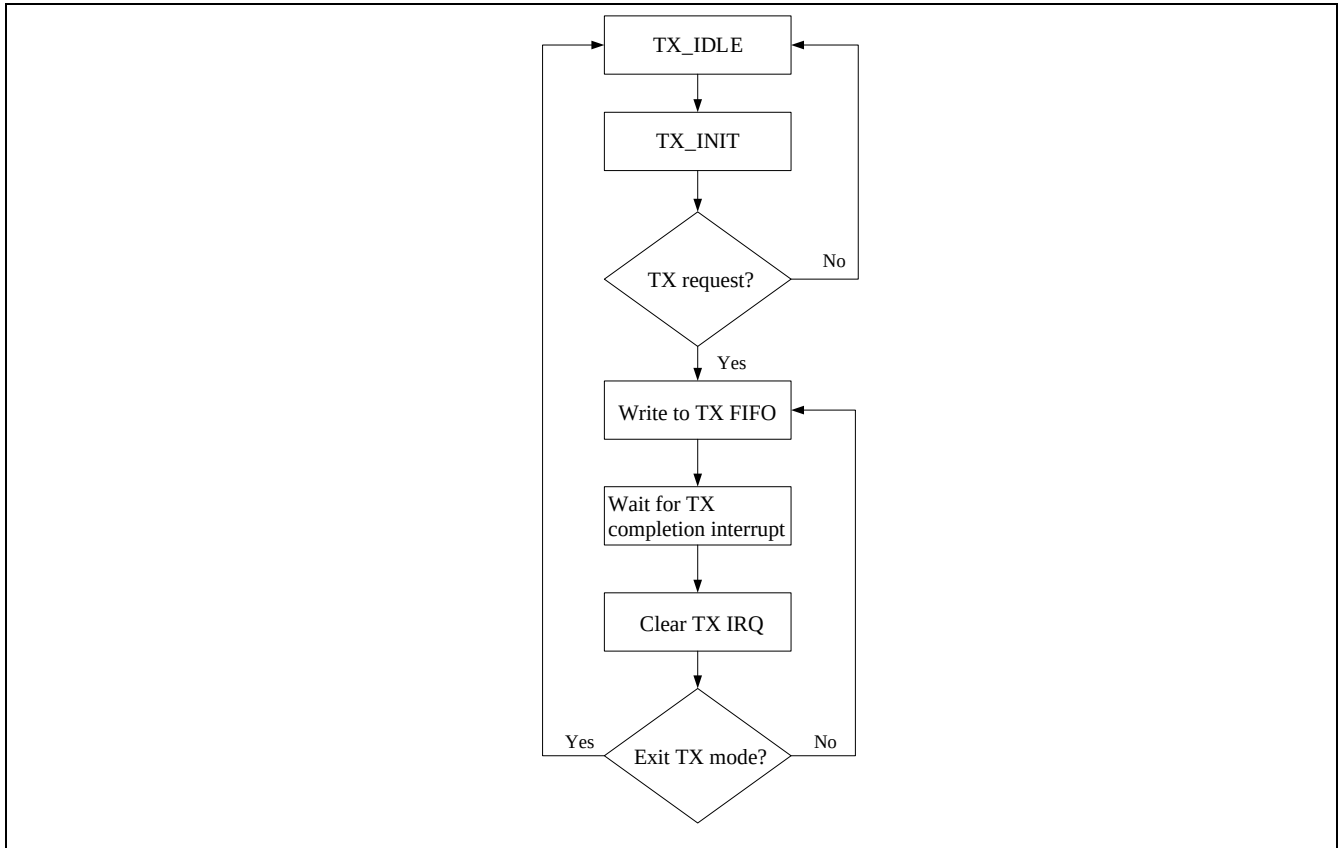


Figure 8-2 Continuous Transmit Mode Flowchart

8.3 Receiver Mode

The MAC receive mode includes single-packet receive, single-packet receive with timeout, and continuous receive modes, selected using a 2-bit mode select register.

When the MAC enters receive mode, the power control module successively turns on analog circuits such as the LDO and PLL, and simultaneously sends an RX start signal to the modem to begin receiving data.

After data reception is complete, the LDO, PLL, and other analog circuits are sequentially turned off. An IRQ signal is then sent to the MCU. After the MCU clears the IRQ, the REG_OPT_MODE register can be configured to enter STB3 mode, completing the RX process.

8.3.1 Single Packet Receive Mode

The state flow chart of the single packet receiving mode is shown in Figure 8-3.

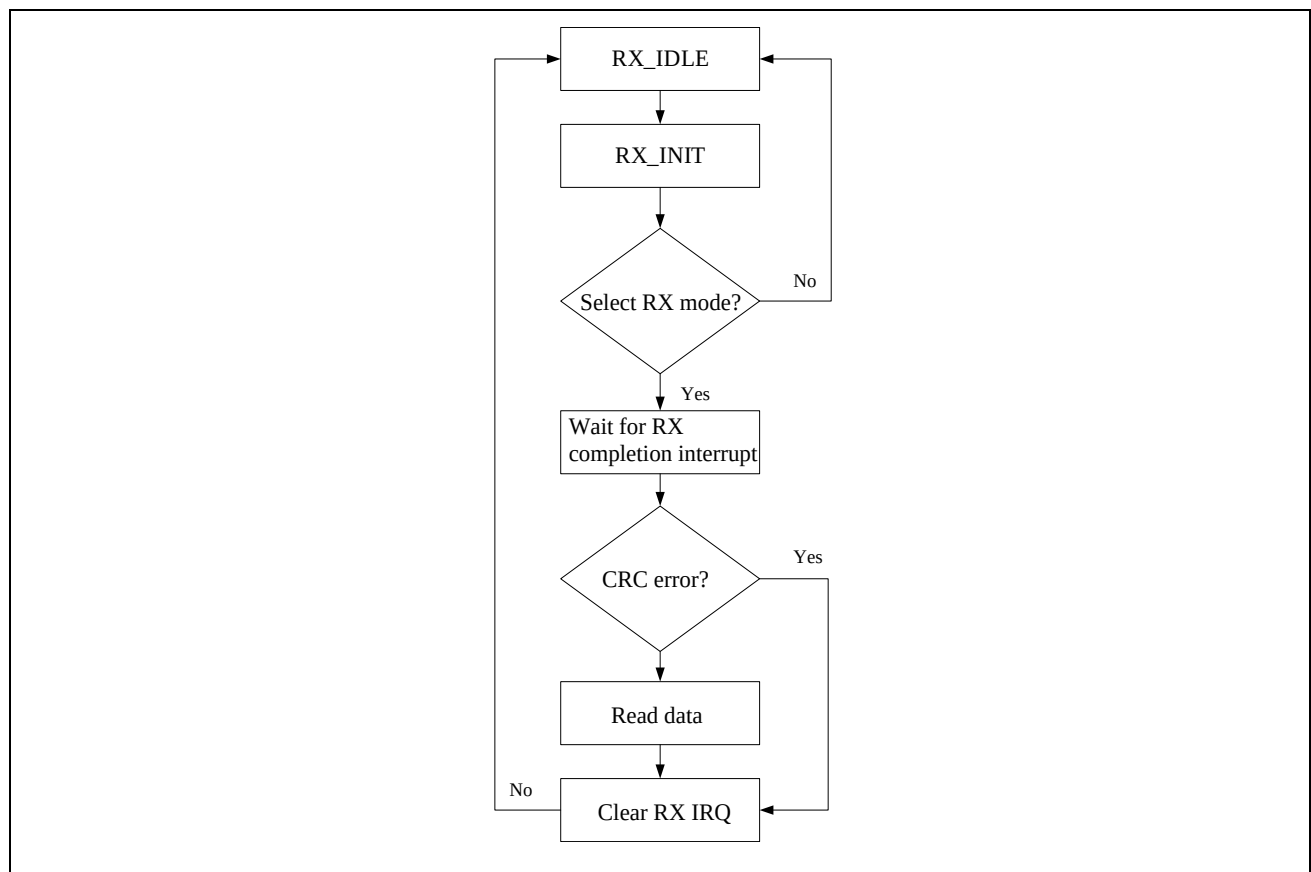


Figure 8-3 Single-Packet Receive Mode Flowchart

8.3.2 Single Packet Receive Mode with Timeout

Figure 8-4 shows the state flow diagram for single-packet receive mode with timeout.

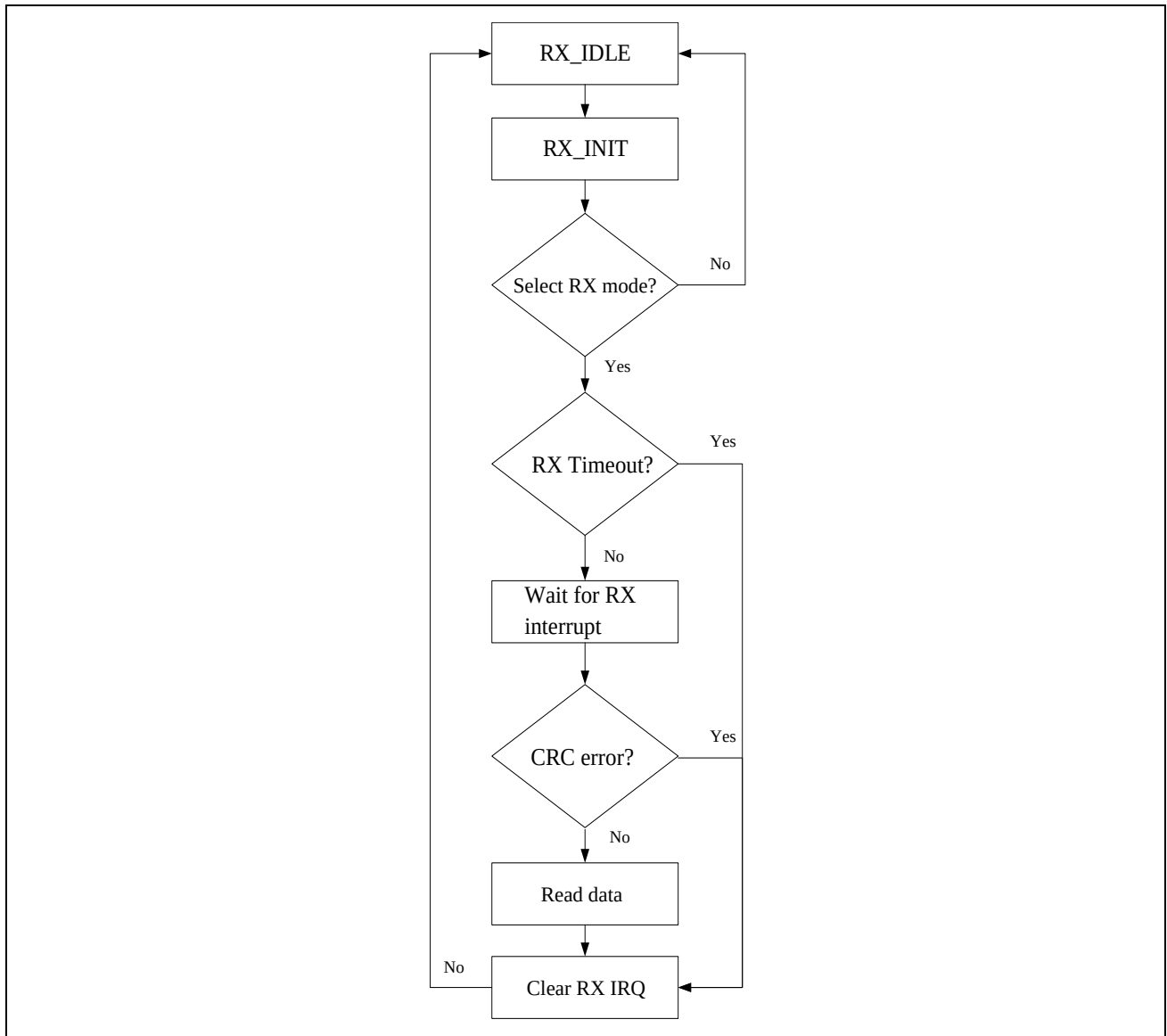


Figure 8-4 Single-Packet Receive Mode with Timeout Flowchart

8.3.3 Continuous Receive Mode

The state flowchart for continuous reception mode is shown in Figure 8-5.

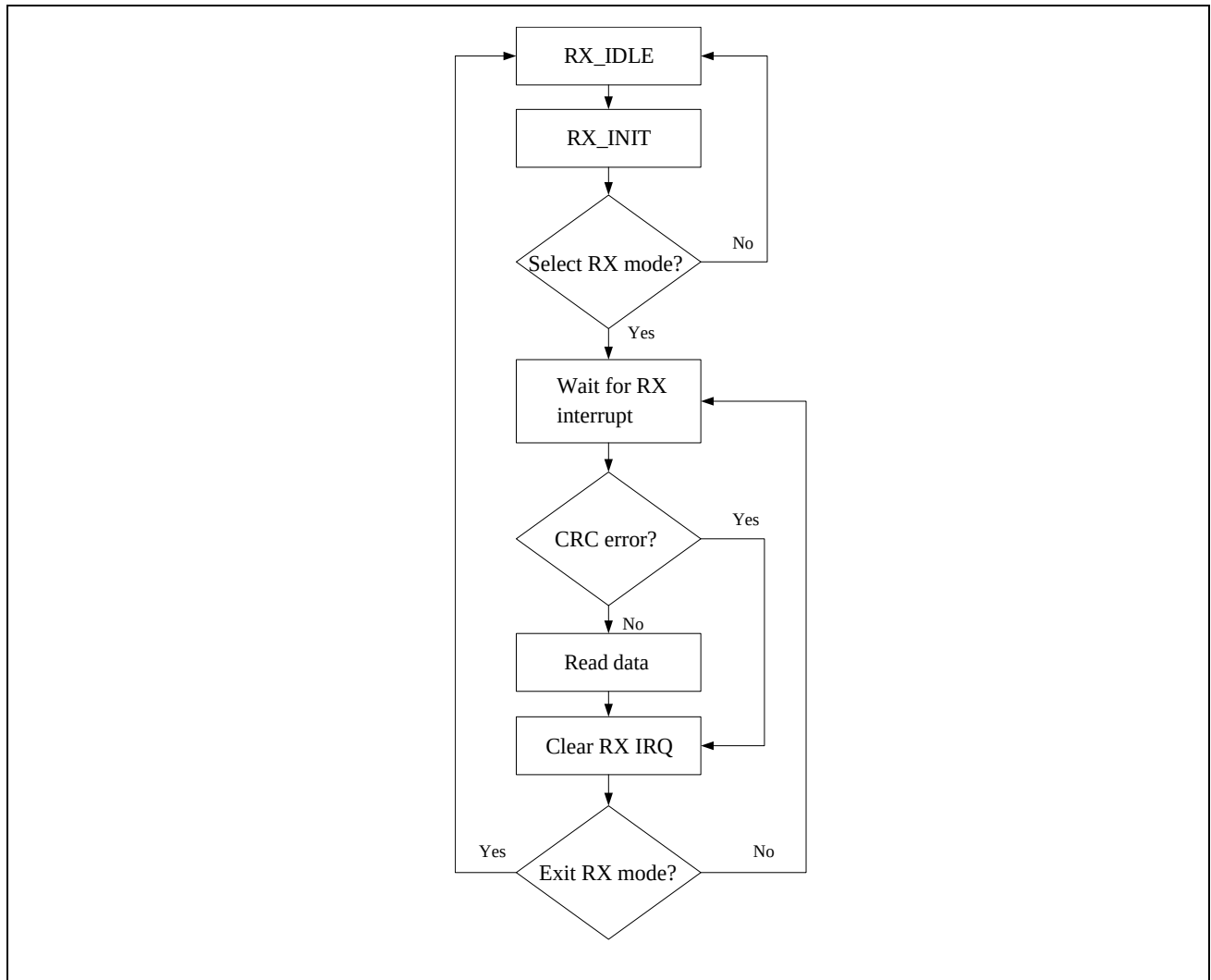


Figure 8-5 Continuous Receive Mode Flowchart

Continuous receive mode differs from single-packet receive mode and single-packet receive mode with timeout in that after data reception is complete, an IRQ signal is sent to the MCU. After the MCU clears the IRQ, the next data reception cycle begins. To exit continuous receive mode, the mode select register selects crystal oscillator output mode. Analog modules such as the LDO and PLL are then disabled, completing the continuous receive mode cycle.

8.4 IRQ Interrupt

All interrupts on this chip share a single external interrupt IO pin, IRQ, which is active high. The chip has seven interrupt types, as listed below. Interrupts can be enabled using the interrupt enable register, and their status can be read using the interrupt status register.

Interrupt Enable Register	Description
REG_MAPM_IRQ_MASK	IRQ Mask Enable in MAPM Mode 0: Generates MAPM IRQ interrupts 1: Does not generate MAPM IRQ interrupts
REG_RX_HMP_ERR_MASK	ECC Error IRQ Interrupt Mask 0: Does not generate IRQ interrupts for ECC errors 1: Generates IRQ interrupts for ECC errors
REG_RX_PLHD_DONE_MASK	Early Interrupt Mode Mask 0: Does not generate IRQ interrupts for early interrupt mode 1: Generates IRQ interrupts for early interrupt mode.
REG_RX_DONE_MASK	RX Completion IRQ Mask 0: Does not generate IRQ interrupts for RX completion 1: Generates IRQ interrupts for RX completion
REG_RX_PL_CRC_ERR_MASK	RX Payload CRC Error IRQ Mask 0: Does not generate IRQ interrupts for RX CRC errors 1: Generates IRQ interrupts for RX CRC errors.
REG_RX_TIMEOUT_MASK	RX Timeout IRQ Mask 0: Does not generate IRQ interrupts for RX timeouts 1: Generates an IRQ interrupt for Rx timeout
REG_TX_DONE_MASK	Tx completion IRQ mask 0: Does not generate an IRQ interrupt for Tx completion 1: Generates an IRQ interrupt for Tx completion

Interrupt Status Register	Description
REG_MDM_MAPM_IRQ	MAPM IRQ interrupt
REG_RX_HM_P_ERR_IRQ	ECC error IRQ interrupt
REG_RX_PLHD_DONE_IRQ	Early interrupt mode IRQ interrupt
REG_RX_DONE_IRQ	RX completion IRQ interrupt
REG_RX_PL_CRC_ERR_IRQ	RX payload CRC error IRQ interrupt
REG_RX_TIMEOUT_IRQ	RX timeout IRQ interrupt
REG_TX_DONE_IRQ	TX completion IRQ interrupt

8.5 CAD-IRQ Interrupt

The chip's CAD-IRQ interrupt pin reuses GPIO11. To use it, you need to configure GPIO11 hardware to CAD-IRQ output mode.

Register Name	Register Location	Description
REG_OUT_FUNC_SEL_CAD	page0, 0x5e, bit6	CAD Function Selection When REG_OUT_FUNC_SEL_CAD is 1, the GPIO functions as a normal IO pin, and its output can be configured using the PAD_OUT_REG register. When REG_OUT_FUNC_SEL_CAD is 0, the GPIO output functions as a CAD pin.
PAD_OUT_REG_CAD	page0, 0x68, bit3	Controls the GPIO11 output register.

9 Interface Design

The chip supports reading and writing registers and transmit/receive FIFOs via four-wire SPI, three-wire SPI, and I2C. By default, four-wire SPI and I2C are both supported. Three-wire SPI requires configuration via the SPI3_EN_REG register.

Note: The address is a special area. The register address in this area is the entry address for this special area, and its operation differs from that of regular registers. Please refer to the section for details on the operation.

9.1 Four-wire SPI

The chip implements an SPI bus slave, used to read and write registers and FIFOs. The SPI bus is a four-wire system consisting of:

- SCK (clock)
- CSN (chip select, active low)
- MOSI (data input)
- MISO (data output)

SCK, CSN, and MOSI are controlled by the master, while MISO is controlled by the slave.

Communication begins when CSN is pulled low and ends when CSN is pulled high. The master sends data via MOSI and receives data via MISO. Data is generated on the falling edge of SCK and sampled on the rising edge.

The information transmitted by the master consists of two parts: the address byte and the data byte. The first 7 bits of the address byte are the address bit (addr); the last bit is the read/write bit (wr). This bit is set to 1 during a write operation and to 0 during a read operation.

SPI has three transmission modes:

Single: Single-byte transmission mode. The message is only 2 bytes long. The master sends the address byte via MOSI. If it is a write operation, the master continues to send data bytes via MOSI. If it is a read operation, the master reads the data bytes replied by the slave on MISO.

Burst: Burst continuous transmission mode. When the message is longer than 2 bytes, the address byte is followed by several data bytes. There is no need to add address bytes between data bytes. The slave automatically increments the address between each data byte. The CSN signal is pulled high after the last data byte and remains low during the rest of the transmission.

FIFO: FIFO read/write mode. This mode supports both single-byte and continuous transmission. The transmission rules are the same as those in Single and Burst modes. The difference is that the address bit addr in the address byte can only be configured as 0x01, and the slave does not increment the address between data bytes.

In addition, the REG_OUT_PAD_MODE register can be configured to set MISO to input high-impedance state to implement a 4-wire SPI one-master-multiple-slave connection architecture.

REG_OUT_PAD_MODE	0 (dec)	1 (dec)	2 (dec)	3 (dec)	3 (dec)
MISO_OUT	MISO_3V	0	1	CSN_IN=1 Output 0	CSN_IN=0 Output MISO_3V
MISO_DIEN	0	0	0	0	0
MISO_OE	1	1	1	CSN_IN=1 0	CSN_IN=0 1
MISO_PUEN	0	0	0	0	0
MISO_PDEN	0	0	0	0	0
NOTE	Normal output	Output 0	Output 1	CSN_IN=1 No output	CSN_IN=0 Normal output

9.1.1 Four-wire SPI write timing

The SPI write timing is as follows:

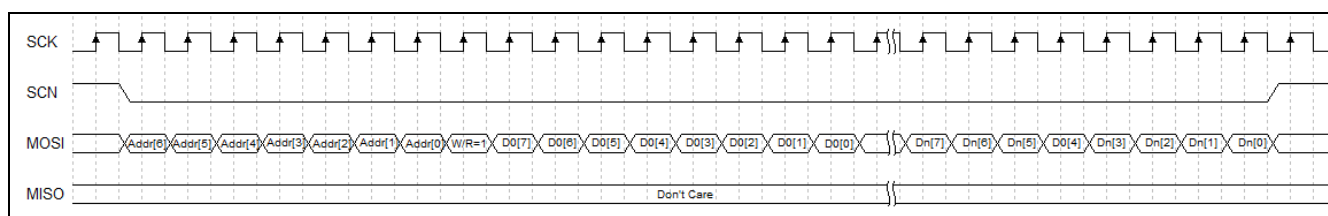


Figure 9-1 SPI Write Timing

9.1.2 Four-wire SPI read timing

The SPI read timing is as follows:

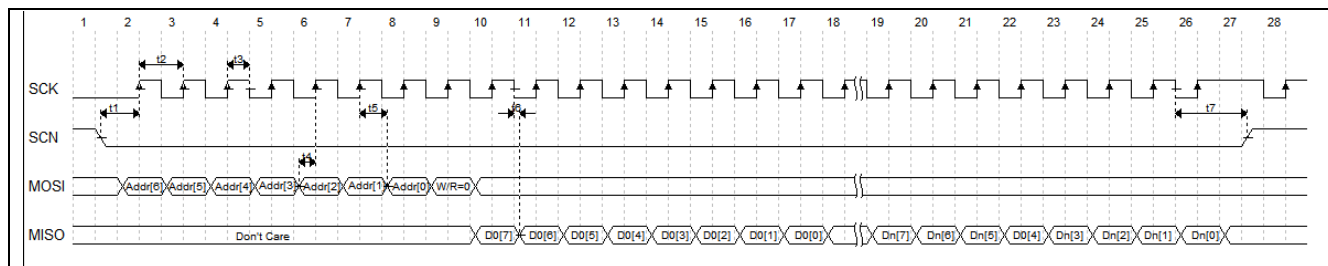


Figure 9-2 SPI Read Timing

9.1.3 Four-wire SPI timing requirements

All SPI timing requirements in the table below are as follows:

Symbol	Description	Min	Typ	Max	Unit
t1	SCN falling edge to SCK rising edge	32	-	-	ns
t2	SCK clock period	100	-	-	ns
t3	SCK high time	50	-	-	ns
t4	MOSI to SCK clock setup time	7	-	-	ns
t5	MOSI to SCK clock hold time	7	-	-	ns
t6	SCK falling edge to MISO delay	0	-	15	ns
t7	SCK to SCN rising edge time	0	-	150	ns

9.2 Three-wire SPI

By default, the chip supports four-wire SPI and I2C. If you require three-wire SPI, you must configure the SPI3_EN_REG register after power-up to enable three-wire SPI. Three-wire SPI signals include:

- SCK (clock)
- CSN (chip select signal, active low)
- MOSI (data input/output)

9.2.1 Three-wire SPI write timing

SPI 写时序如下:

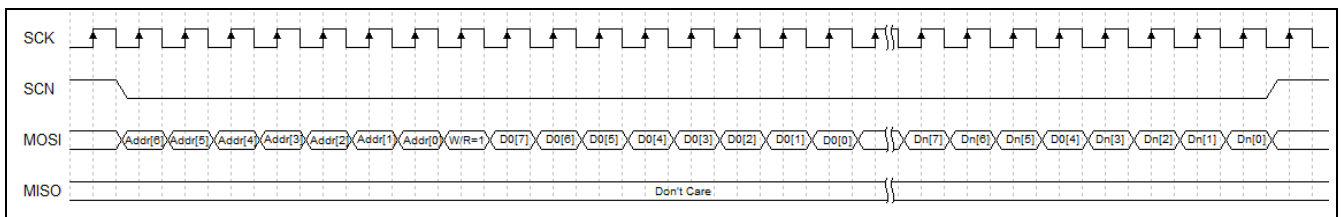


Figure 9-3 3-Wire SPI Write Timing

9.2.2 Three-wire SPI read timing

The SPI write timing is as follows:

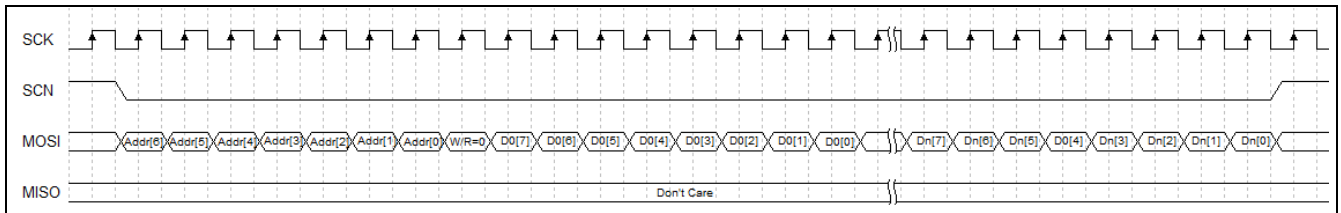


Figure 9-4 3-Wire SPI Read Timing

9.2.3 Three-wire SPI timing requirements

Please refer to Section 9.1.3.

9.3 I2C

The chip supports I2C by default. The I2C signals SCL and SDA multiplex the SPI signals SCK and MOSI.

Note: When using I2C, the CSN signal must be high.

The I2C device address is 7 bits wide and contains 0x72.

The timing diagram of the I2C start signal is as follows:

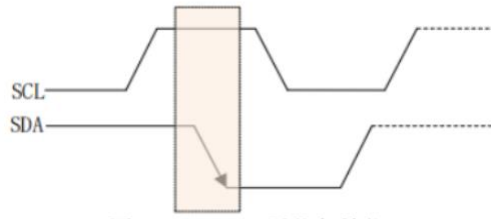


Figure 9-5 I2C Start Signal

The I2C stop signal timing diagram is as follows:

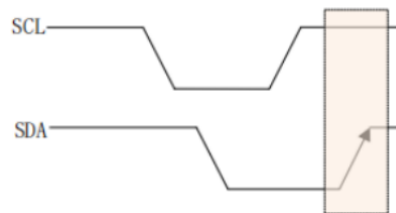


Figure 9-6 I2C Stop Signal

9.3.1 I2C Write Timing

The I2C write timing is as follows:

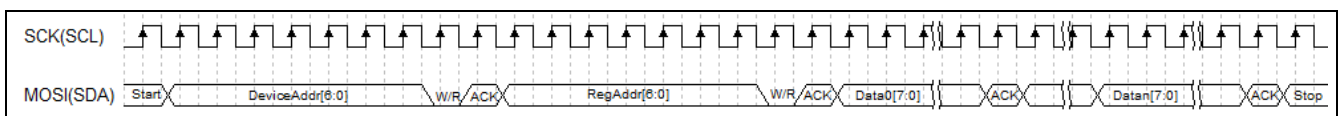


Figure 9-7 I2C Write Timing

9.3.2 I2C Read Timing

The I2C read timing is as follows:

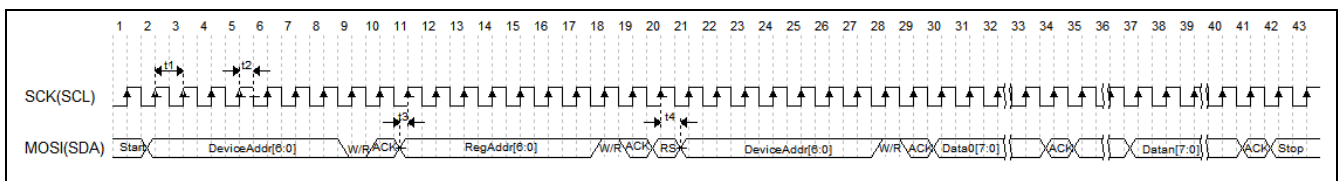


Figure 9-8 I2C Read Timing

9.3.3 I2C Timing Requirements

Symbol	Description	Min	Typ	Max	Unit
t1	SCL clock period	200	-	-	ns
t2	SCL high time	100	-	-	ns
t3	SDA to SCL clock setup time	12	-	-	ns
t4	SDA to SCL clock hold time	12	-	-	ns

9.4 FIFO

The chip features a 256-byte FIFO for storing data transmitted by the TX module and decoded data from the RX module.

The FIFO consists of a single-port RAM and can only store and read a single packet of data.

If a packet already exists in the FIFO, it must be read before writing to the previous packet; otherwise, the previous packet in the FIFO will be overwritten.

In STB3 and later operating modes, the FIFO can be read and written via a modem, SPI, or I2C.

The FIFO's read and write address is 0x01, which can be used to read and write the FIFO contents via SPI or I2C.

9.5 GPIO

The chip supports multiple GPIO input and output control, which can be controlled through register configuration. For register configuration, please refer to Section 12.4. Each GPIO port can be configured as an internal pull-up/pull-down input, a high-impedance input (floating input), or a push-pull output (CMOS output). The configuration methods are shown in Table 9-1.

Table 9-1 GPIO port mode configuration

DIEN	OE	OUT	PUEN	PDEN	IO Direction	IO Status
0	1	0/1	0	0	Push-Pull Output	0/1
0	0	x	0	0	Open-Drain Output	High-Impedance
1	0	x	0	0	Input	High-Impedance
1	0	x	1	0	Input	Pull-Up
1	0	x	0	1	Input	Pull-Down

Note: The output modes of GPIO10 and GPIO11 in the chip can be reused to implement special functions.

9.5.1 External PA Control GPIO

GPIO10 has two configurable output modes: normal IO and PA control pin. GPIO10's different modes can be configured using the REG_OUT_FUNC_SEL_PA register.

When GPIO10 is configured as a normal IO, it can output high or low levels, just like any other GPIO.

When GPIO10 is configured as a PA control pin and the external PA control mode is enabled via the register, GPIO10 functions as an external PA enable control pin. That is, GPIO10 outputs a high level only when transmitting data in transmit mode. It outputs a low level at all other times.

Register Name	Register Location	Description
REG_OUT_FUNC_SEL_PA	page0, 0x5e, bit7	GPIO10 Function Select When REG_OUT_FUNC_SEL_PA is 1, GPIO10 is a normal I/O pin, and its output can be configured using the PAD_OUT_REG_PA register. When REG_OUT_FUNC_SEL_PA is 0, GPIO10 output functions as an external PA pin.
PAD_OUT_REG_PA	page0, 0x68, bit2	Register that controls the GPIO10 output. When REG_OUT_FUNC_SEL_PA is 1, GPIO10 is a normal I/O pin, and its output can be configured using the PAD_OUT_REG_PA register.

9.5.2 CAD monitoring GPIO

GPIO11 has two configuration modes: normal IO and CAD detection pin. The REG_OUT_FUNC_SEL_CAD register configures the different modes of GPIO11.

When configured as a normal IO, GPIO11 can output high and low levels, just like other GPIOs.

When GPIO11 is configured as a CAD signal detection interrupt output, the CAD function is enabled via registers, and the chip is in RX mode, upon receiving a signal with the configured bandwidth and spreading factor, GPIO11 generates a continuous, stable high level, indicating that the chip has detected an airborne signal. This high level remains low until the signal disappears. In all other situations and modes, GPIO11 outputs a low level. For detailed CAD configuration and usage, please refer to the section.

The chip's CAD-IRQ interrupt pin reuses the GPIO11 pin. When using it, you need to configure the GPIO11 hardware to CAD-IRQ output mode.

Register Name	Register Location	Description
REG_OUT_FUNC_SEL_CAD	page0, 0x5e, bit6	CAD function selection When REG_OUT_FUNC_SEL_CAD is 1, the GPIO is a normal I/O pin, and its output can be configured using the PAD_OUT_REG register. When REG_OUT_FUNC_SEL_CAD is 0, the GPIO output is a CAD function pin.
PAD_OUT_REG_CAD	page0, 0x68, bit3	Register that controls the output of GPIO11. When REG_OUT_FUNC_SEL_CAD is 1, GPIO11 is a normal I/O pin, and its output can be configured using the PAD_OUT_REG_CAD register.

10 ChirpIoT special function description

10.1 RSSI Function

10.1.1 RSSI indication of ChirpIoT signal

The chip supports signal energy statistics. When receiving ChirpIoT data, the chip's internal logic calculates the RSSI value of the current packet's signal strength. After receiving the packet, the MCU can read the RSSI_R32_HOLD or RSSI_RBW_HOLD registers via the SPI interface to obtain the RSSI value for the current packet.

10.1.2 Channel detection RSSI indication

The RSSI function also supports channel energy statistics during detection. After the chip enters the RX state, RSSI energy statistics can be performed by configuring the REG_RSSI_PRE_RD register. The current signal RSSI energy can then be read using the RSSI_R32_HOLD register.

Configuring REG_RSSI_PRE_RD to 0 and then to 1 updates the RSSI value. The register location is as follows:

Register Name	Register Location	Description
REG_RSSI_PRE_RD	page2, 0x06, bit2	Detects the channel RSSI reading switch. Configuring from 0 to 1 updates the RSSI value.

10.2 Signal-Noise Energy Statistics Function

The chip supports signal-noise statistics. When the chip receives ChirpIoT data, its internal logic calculates the signal energy and noise energy of the current packet. After receiving the packet, the MCU can read the signal energy value in registers SIG_POW_AVG and the noise energy value in register NOI_POW_AVG via the SPI interface. The MCU can then calculate the signal-to-noise ratio of the packet using a standard formula.

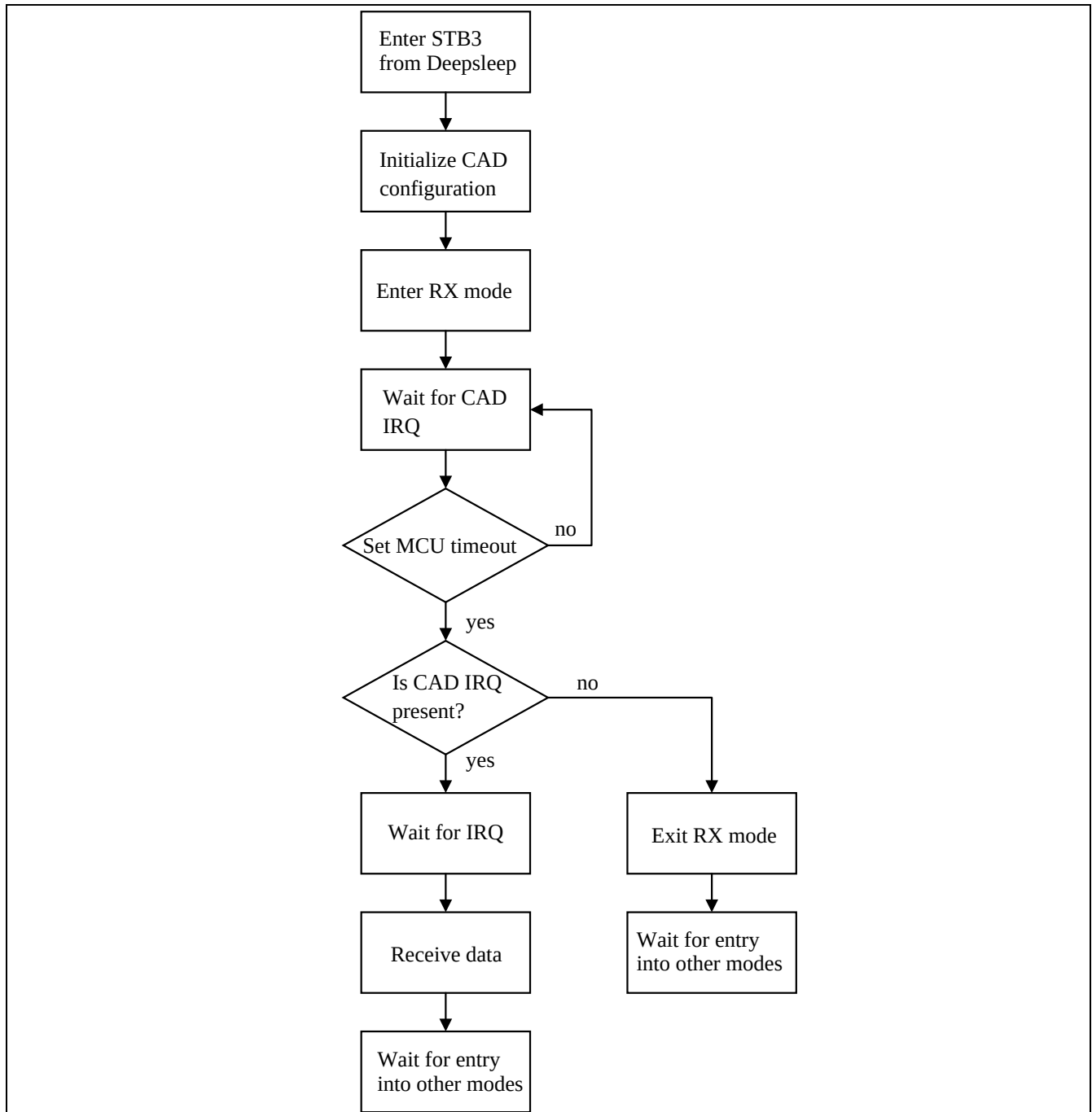
10.3 Channel Activity Detection (CAD)

Because ChirpIoT™ signals have excellent reception performance and can still be properly received and demodulated even when their signal strength is below the receiver's noise floor, using only the RSSI (signal strength indicator) to determine channel occupancy is insufficient. Therefore, the chip has a dedicated channel activity detector to detect ChirpIoT™ signals. This ensures that even when the signal strength is below the receiver's noise floor, it can detect channel occupancy by ChirpIoT™ signals from other devices.

The detection function of the channel interaction detector is divided into two parts: one is the ChirpIoT™ signal preamble detection, and the other is the ChirpIoT™ signal data segment detection.

10.3.1 ChirpIoT signal preamble detection

As shown in Figure 10-1:



The chip supports CAD-IRQ interrupts. After entering RX mode, it detects whether a ChirpIoT™ preamble signal is present in the channel. If so, it sets CAD-IRQ high. An external MCU can determine whether the ChirpIoT™ preamble signal is present by monitoring whether the CAD-IRQ signal remains high for a specified period of time. If so, it waits for the data receive IRQ interrupt and proceeds with data reception. If no ChirpIoT™ preamble signal is present, it quickly exits RX mode and awaits a command to enter another mode.

In addition, to reduce false CAD-IRQ triggering, a CAD-IRQ filtering function has been added, which can be enabled via a configuration register.

Address	Register	BIT	Default Value	Read/Write	Description
[1][0x25]	REG_DEC_VAL_THRE	1:0	0x3	W/R	CAD detection threshold, which specifies the number of consecutive preambles received before CAD is pulled high.

10.3.2 ChirpIoT signal data segment detection

In order to avoid missing the preamble of the ChirpIoT signal and thus failing to perform effective CAD detection during the actual reception process, the chip also supports the detection of non-ChirpIoT preamble signals, namely the ChirpIoT data segment signal detection function.

The data segment signal detection can be enabled by configuring the registers REG_SIGNAL_POWER_THRED, REG_DEC_PLD_THRE, and REG_DEC_PLD_EN.

Register Configuration Description:

Address	Register	BIT	Default Value	Read/Write	Description
[1][0x35]	REG_SIGNAL_POWER_THRED	7:4	4'b1110	W/R	CAD valid signal threshold. Only when the energy exceeds this value will the CAD signal be considered valid.
	REG_DEC_PLD_THRE	3:2	2'b10	W/R	CAD valid signal count threshold. Only when this number of preambles meeting the condition is received will the CAD signal be raised.
	REG_DEC_PLD_EN	1	1'b0	W/R	CAD payload detection enable 0: disable 1:enable

Note: CAD detection for both the preamble and data segments can be enabled simultaneously to ensure that data at different stages can be detected and checked.

10.4 MAPM Function

In ChirpIoT star networking scenarios, to ensure that node devices can receive data sent by the gateway, the gateway uses an extra-long preamble, such as one second. In this scenario, all node devices, after receiving the preamble, will continue receiving until the payload phase displays the node's network address. At this point, the node device's software can compare the received network address with its own address. If they differ, reception stops, reducing power consumption. Reception resumes if the address reception conditions are met. This solution presents several issues:

- ① Node devices that do not meet the address reception conditions cannot stop receiving until the payload phase, wasting significant power.
- ② Node devices that meet the address reception conditions also need to receive the entire extra-long preamble before entering the payload phase, which also wastes power.

To address these two power consumption issues, the chip adds a new frame structure mode, Multiple Address Preamble Mode (MAPM). This frame structure inserts a special address into the preamble phase. This allows nodes to preemptively determine whether they meet the address reception requirements based on the address, thus resolving issue 1.

Furthermore, to address issue 2, a variable counter is added to the special address to indicate the remaining preamble length.

Nodes that meet the address reception requirements can enter a low-power sleep state in advance. This counter can then be used to determine how long in advance they need to enter the receive state to ensure normal data reception, further reducing power consumption for nodes that meet the address reception requirements.

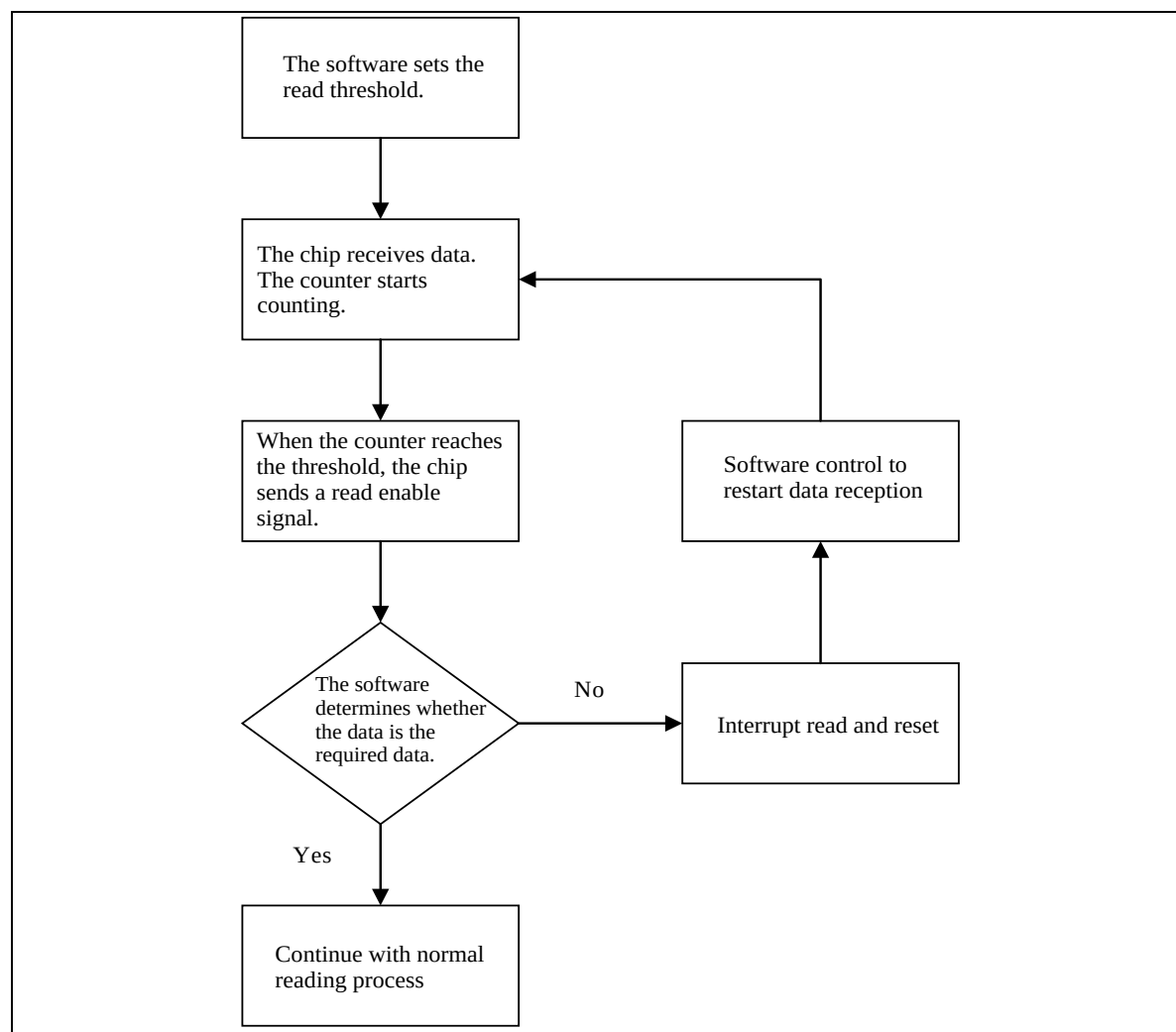
10.5 eFuse Function

The chip supports 82 bytes of eFuse space for customers to use, which can be used to store data that will not be lost during power failure.

10.6 Pre-interrupt Function

The chip's early interrupt function allows the chip to check the decoded data while reading a frame of data, determine whether it is what it wants, and then decide whether to continue reading or abandon the frame.

The flow chart is as follows:



10.7 Intelligent SF Identification

To achieve lightweight gateway functionality, the chip incorporates intelligent SF recognition hardware circuitry. This circuit intelligently identifies the ChirpIoT™ SF pattern in the actual channel at the same frequency and automatically configures the chip's SF parameters to receive data from signals with different SFs.

The specific implementation process is shown in Figure 10-2:

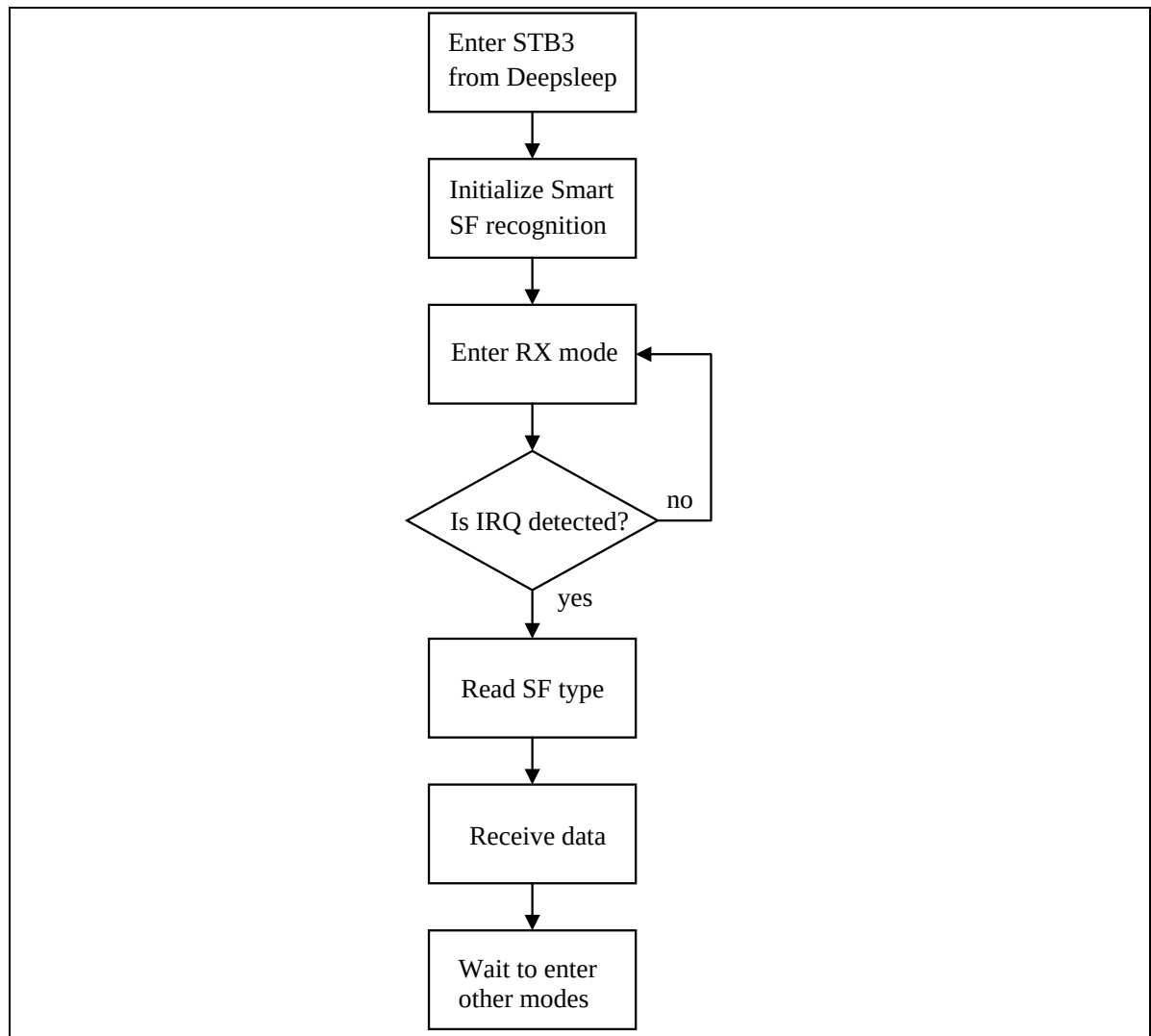


Figure 10-2 Intelligent SF Identification

11 Operation Modes Description

11.1 Overview

The chip has 7 modes: DeepSleep mode, Sleep mode, STB1 mode, STB2 mode, STB3 mode, TX mode and RX mode.

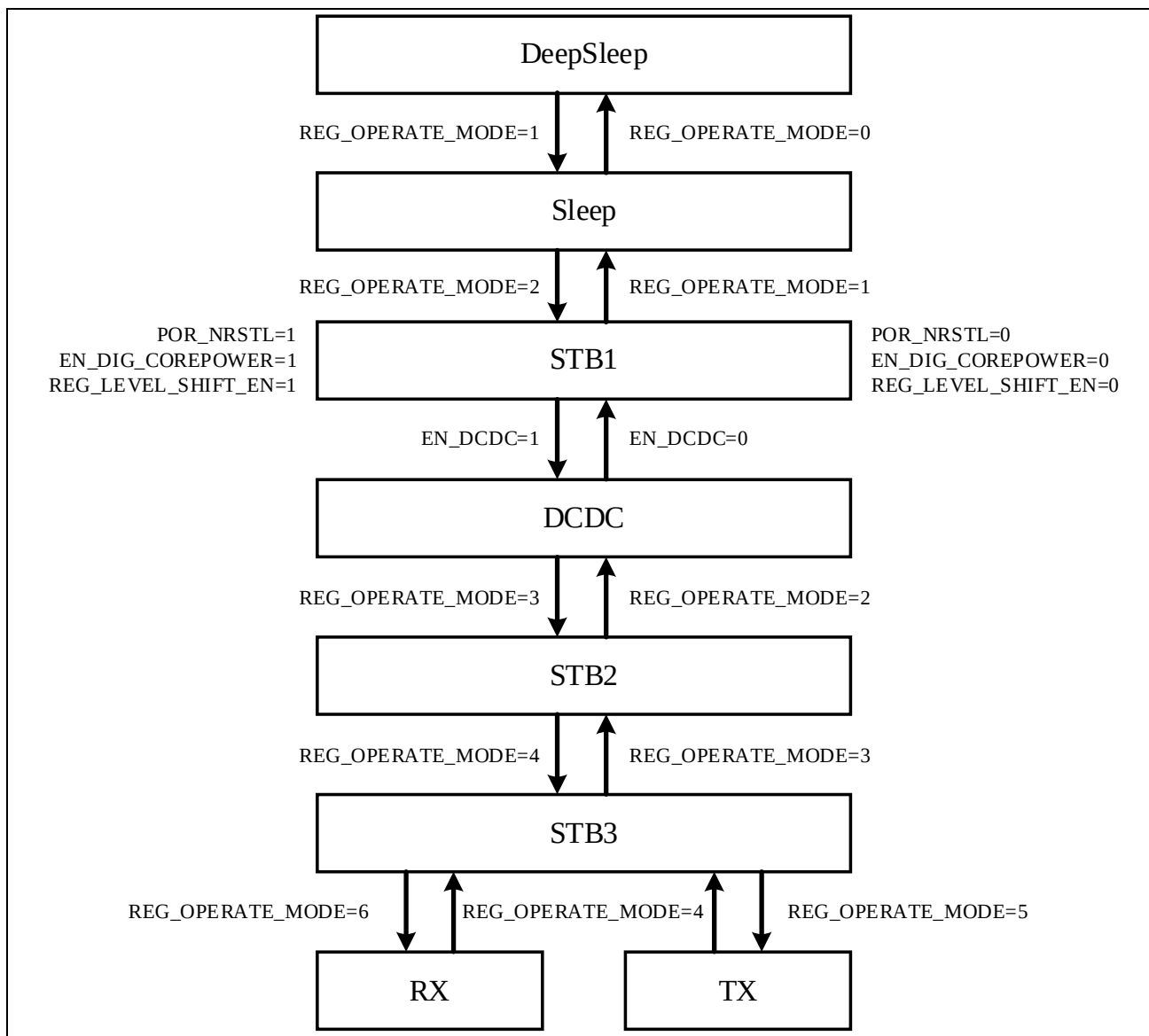


Figure 11-1 Operating Mode Setting Process

After power-on, the chip has seven states: Deep Sleep, Sleep, LDO Operation (STB1), OSC Operation (STB2), OSC Output (STB3), TX, and RX. All states can be switched via configuration registers.

Note: If the chip's peripheral circuitry does not support DCDC, EN_DCDC can be omitted and the chip can be switched directly from STB1 to STB2. For chip circuitry, refer to Chapter 13.

11.2 Working Modes Description

11.2.1 Deep Sleep Mode

In this mode, the 3V logic area operates. Except for registers in the 3V logic area, which are configurable, other modules are inoperable, minimizing chip power consumption.

11.2.2 Sleep Mode

In this mode, the low-power LDO is enabled, resulting in lower chip power consumption and maintaining power for all registers. However, SPI/I2C is disabled for registers other than those in the 3V logic area.

11.2.3 STB1 Mode

In this mode, the high-performance LDO begins operating and the low-voltage area logic is enabled.

11.2.4 DCDC Mode

In STB1 mode, the DCDC function can be enabled by writing a high level to the EN_DCDC register.

Note: DCDC mode requires peripheral circuit support; please refer to Chapter 13.

11.2.5 STB2 Mode

In this mode, the OSC crystal oscillator is enabled. The OSC starts operating and maintains oscillation, but does not output to other modules, resulting in relatively low power consumption.

11.2.6 STB3 Mode

In this mode, the OSC clock is output to all modules, and the chip begins normal operation.

11.2.7 TX Mode

In this mode, all TX transmit-related modules are enabled, enabling packet data encoding and transmission.

11.2.8 RX Mode

In this mode, all RX receive-related modules are enabled, awaiting the reception and decoding of packet data.

12 Registers

The chip's registers are organized into four pages, each containing 128 register addresses. Register page selection is performed using the REG_PAGE_SEL bit in register 0x00. Registers at addresses 0x00 to 0x05 can be accessed from any register page. The following table describes the location and function of each register on the chip.

Note:

- Read and write operations are prohibited on registers in the Reserved section.
- All registers in the low-voltage area are only supported in STB1, STB2, STB3, Tx, and Rx modes.

12.1 System Configuration

Page	Address	Register	BIT	Default Value	Read/Write	Description
0/1/2/3	0x00	REG_SOFT_RST	7	0x00	W/R	Soft reset, all except 1.2V configuration registers 1.2V logic reset 0: Reset 1: Do not reset
		Reserved	6:2		W/R	Reserved
		REG_PAGE_SEL	1:0		W/R	0: Select Page 0 1: Select Page 1 2: Select Page 2 3: Select Page 3
	0x01	FIFO READ/ WRITE ACCESS POINT	7:0	-	W/R	FIFO read and write address

12.2 Mode Configuration

Page	Address	Register	BIT	Default Value	Read/Write	Description
0/1/2/3	0x02	Reserved	7:3	0x00	W/R	Reserved
		REG_OPERATE_MODE	2:0		W/R	0: Deep Sleep Mode 1: Sleep Mode 2: STB1 Mode (LDO Operation) 3: STB2 Mode (OSC Operation) 4: STB3 Mode (OSC Output) 5: Tx Mode 6: Rx Mode
	0x04	Reserved	7:6	0x06	W/R	Reserved
		EN_LS_3V	5		W/R	1: Level Shift Enable
		POR_RSTL	4		W/R	Reset All 1.2V Configuration Registers 0: Reset 1: Do Not Reset
		Reserved	3:0		W/R	Reserved

12.3 MAC Configuration (Low Voltage Area Register)

Page	Address	Register	BIT	Default Value	Read/Write	Description
0	0x54	Reserved	7:3	0x00	W/R	Reserved
		REG_TX_FIFO_FULL	2		W/R	0: TX FIFO is not full 1: TX FIFO is full
		REG_TX_FIFO_FULL_SEL	1		W/R	0: select fifo wr done 1: select reg tx fifo full
		Reserved	0		W/R	Reserved
	0x58	REG_RX_TIMEOUT_IMMED	7	0x0F	W/R	1: Generate an IRQ immediately for an Rx timeout
		REG_MAPM_IRQ_MASK	6		W/R	IRQ mask for MAPM mode 0: Interrupt disabled 1: Interrupt enabled
		Reserved	5		W/R	Reserved
		REG_RX_PLHD_DONE_MASK	4		W/R	Early interrupt mode mask 0: Interrupt disabled 1: Interrupt enabled
		REG_RX_DONE_MASK	3		W/R	RX completion IRQ mask 0: Interrupt disabled 1: Interrupt enabled
		REG_RX_PL_CRC_ERR_MASK	2		W/R	Rx payload CRC error IRQ mask 0: Interrupt disabled 1: Interrupt enabled
		REG_RX_TIMEOUT_MASK	1		W/R	Rx timeout IRQ mask 0: Interrupt disabled 1: Interrupt enabled
		REG_TX_DONE_MASK	0		W/R	Tx completion IRQ Mask 0: Interrupt disabled 1: Interrupt enabled
	0x5b	Reserved	7:3	0x03	W/R	Reserved
		REG_RX_DONE_RST_TEST	2		W/R	1: Modem resets immediately upon RX completion 0: Modem resets after IRQ is cleared and RX is completed
		REG_RX_CR_ERR_RST	1		W/R	1: RX rate error reset enable
		REG_RX_HEADER_ER_RST	0		W/R	1: RX header error reset enable
	0x5d	Reserved	7:1	0x00	W/R	Reserved
		REG_2POINT_CAL_EN	0		W/R	Two-point CAL enable, active high
	0x6c	EFUSE_WR_IRQ	7	0x00	R	eFuse Read/Write Completion Interrupt 1: High indicates read/write completion
		REG_MDM_MAPM_IRQ	6		R	MAPM Interrupt Status Register 1: MAPM IRQ; 0: No MAPM IRQ
		Reserved	5		W/R	Reserved
		REG_RX_PLHD_DONE_IRQ	4		W/R	Write 1 to clear the early interrupt
		REG_RX_DONE_IRQ	3		W/R	Write 1 to clear the RX done IRQ
		REG_RX_PL_CRC_ERR_IRQ	2		W/R	Write 1 to clear the RX payload CRC error IRQ

		REG_RX_TIMEOUT_IRQ	1		W/R	Write 1 to clear rx timeout irq
		REG_TX_DONE_IRQ	0		W/R	Write 1 to clear tx done irq
	0x6D	REG_MAPM_PN_CNT	7:0	0x00	R	Preamble count counter for MAPM mode
	0x6E	REG_MAPM_ADDRX	7:0	0x00	R	Addrx for MAPM mode

12.4 GPIO interface configuration (low voltage area register)

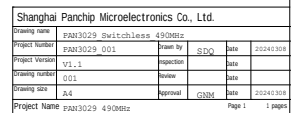
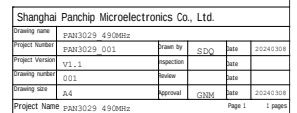
Page	Address	Register	BIT	Default Value	Read/Write	Description
0	0x5e	REG_OUT_FUNC_SEL_PA	7	0x00	W/R	1: Used as normal IO 0: Used for external PA control
		REG_OUT_FUNC_SEL_CAD	6		W/R	1: Used as normal IO 0: Used as CAD pin
		Reserved	5:0		W/R	Reserved
	0x5f	GPIO_PDEN LSB	7:0	0x00	W/R	GPIO_PDEN LSB[3]: GPIO3 pull-down enable GPIO_PDEN LSB[0]: GPIO0 pull-down enable Other configurations Reserved
	0x60	Reserved	7:4	0x00	W/R	Reserved
		GPIO_PDEN MSB	3:0		W/R	GPIO_PDEN MSB[3]: GPIO11 pull-down enable GPIO_PDEN MSB[2]: GPIO10 pull-down enable Other configurations Reserved
	0x61	GPIO_PUEN LSB	7:0	0x00	W/R	GPIO_PDEN LSB[3]: GPIO3 pull-up enable GPIO_PDEN LSB[0]: GPIO0 pull-up enable Other configurations Reserved
	0x62	Reserved	7:4	0x00	W/R	Reserved
		GPIO_PUEN MSB	3:0		W/R	GPIO_PDEN MSB[3]: GPIO11 pull-up enable GPIO_PDEN MSB[2]: GPIO10 pull-up enable Other configurations Reserved
	0x63	GPIO_DIEN LSB	7:0	0x00	W/R	GPIO_PDEN LSB[3]: GPIO3 input enable GPIO_PDEN LSB[0]: GPIO0 input enable Other configurations Reserved
	0x64	Reserved	7:4	0x00	W/R	Reserved
		GPIO_DIEN MSB	3:0		W/R	GPIO_PDEN MSB[3]: GPIO11 input enable GPIO_PDEN MSB[2]: GPIO10 input enable Other configurations Reserved
	0x65	GPIO_OE LSB	7:0	0x00	W/R	GPIO_PDEN LSB[3]: GPIO3 output enable GPIO_PDEN LSB[0]: GPIO0 output enable Other configurations Reserved
	0x66	Reserved	7:4	0x00	W/R	Reserved
		GPIO_OE MSB	3:0		W/R	GPIO_PDEN MSB[3]: GPIO11 output enable GPIO_PDEN MSB[2]: GPIO10 output enable Other configurations Reserved
	0x67	GPIO_OUT LSB	7:0	0x00	W/R	GPIO_PDEN LSB[3]: GPIO3 output GPIO_PDEN LSB[0]: GPIO0 output Other configurations Reserved
	0x68	Reserved	7:4	0x00	W/R	Reserved
		GPIO_OUT MSB	3:0		W/R	GPIO_PDEN MSB[3]: GPIO11 output

						GPIO_PDEN MSB[2]: GPIO10 output Other configurations Reserved
	0x74	Reserved	7:6	-	R	Reserved
		GPIO_IN LSB	5:0	-	R	GPIO_PDEN LSB[3]: GPIO3 input GPIO_PDEN LSB[0]: GPIO0 input Other configurations Reserved
	0x75	Reserved	7:6	-	R	Reserved
		GPIO_IN MSB	5:0	-	R	GPIO_PDEN LSB[3]: GPIO11 input GPIO_PDEN LSB[2]: GPIO10 input Other configurations Reserved

12.5 Basic Operation Configuration

Page	Address	Register	BIT	Default Value	Read/Write	Description
3	0x06	Reserved	7:3	0x00	W/R	Reserved
		REG_TX_CFG_MODE	2		W/R	0: Single transmit mode 1: Continuous transmit mode
		REG_RX_CFG_MODE	1:0		W/R	0: Single receive mode 1: Single receive mode with timeout 2: Continuous receive mode
	0x07	REG_RX_TIMEOUT[7:0]	7:0	0x00	W/R	Receive timeout setting, lower 8 bits, in microseconds
	0x08	REG_RX_TIMEOUT[15:8]	7:0	0x00	W/R	Receive timeout setting, upper 8 bits, in microseconds
	0x0D	BW	7:4	0x98	W/R	Bandwidth: 6: 62.5 kHz 7: 125 kHz 8: 250 kHz 9: 500 kHz
		CODING RATE	3:1		W/R	Coding rate: 1: 4/5 2: 4/6 3: 4/7 4: 4/8
		Reserved	0		W/R	Reserved
	0x0E	SPREADING FACTOR	7:4	0x78	W/R	SF: 5: 32 chips / symbol 6: 64 chips / symbol 7: 128 chips / symbol 8: 256 chips / symbol 9: 512 chips / symbol 10: 1024 chips / symbol 11: 2048 chips / symbol 12: 4096 chips / symbol
		RXPAYLOADCRC CON	3		W/R	0: CRC disabled 1: CRC enabled
		Reserved	2:0		W/R	Reserved
	0x0F	SYNCWORD[7:4]	7:5	0x12	W/R	Sync word high 4 bits

						SF5 value range: 0x1 to 0x3 SF6 value range: 0x1 to 0x7 other SF values range: 0x1 to 0xF
		SYNCWORD[3:0]	3:0		W/R	Sync word lower 4 bits SF5 value range: 0x0 to 0x3 SF6 value range: 0x0 to 0x7 other SF values range: 0x0 to 0xF
	0x10	IF[7:0]	7:0	0xC0	W/R	Intermediate frequency configuration, second highest 8 bits
	0x11	IF[11:8]	7:4	0x0F	W/R	Intermediate frequency configuration, upper 4 bits
		Reserved	3:0		W/R	Reserved
	0x12	Reserved	7:4	0x06	W/R	Reserved
		LOWDATARATE	3		W/R	0: Normal mode 1: Low rate mode
		Reserved	2:0		W/R	Reserved
	0x13	PREAMBLE_LEN[7:0]	7:0	0x08	W/R	preamble length lower 8 bits
	0x14	PREAMBLE_LEN[15:8]	7:0	0x00	W/R	preamble length higher 8 bits
	0x1A	SPI3_EN_REG	7	0x03	W/R	3-wire SPI Select Register 0: 3-wire SPI mode is not selected 1: 3-wire SPI mode is selected
		REG_MOSI_PUEN	6		W/R	Register control mode for the MOSI pin (PUEN control register)
		REG_CSN_PUEN	5		W/R	Register control mode for the CSN pin (CSN control register)
		REG_SCK_PUEN	4		W/R	Register control mode for the SCK pin (SCK control register)
		REG_IN_PAD_MODE	3		W/R	Enable manual configuration mode for the SPI input pins
		REG_OUT_PAD_MODE	2:0		W/R	SPI MISO pin mode selection, used for multipoint SPI applications
	0x24	Reserved	7:4	0x00	W/R	Reserved
		EN_DCDC_3V	3		W/R	DCDC Enable 0: DCDC Bypass 1: DCDC Operation
		Reserved	2:0		W/R	Reserved
	0x26	XTAL_ACTIVE_EN_3V	7	0x00	W/R	Enables the active crystal oscillator channel that supports 3V power supply, ensuring the chip can operate with the crystal oscillator's 3V clock. 1: On; 0: Off
		XTAL_STARTUP_EN_3V	6		W/R	Crystal Oscillator Fast Startup Enable. 1: On; 0: Off.
		EN_DIG_COREPOWER_3V	5		W/R	Digital Core Voltage Enable Switch 1: On; 0: Off
		Reserved	4:0		W/R	Reserved



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14 Package size

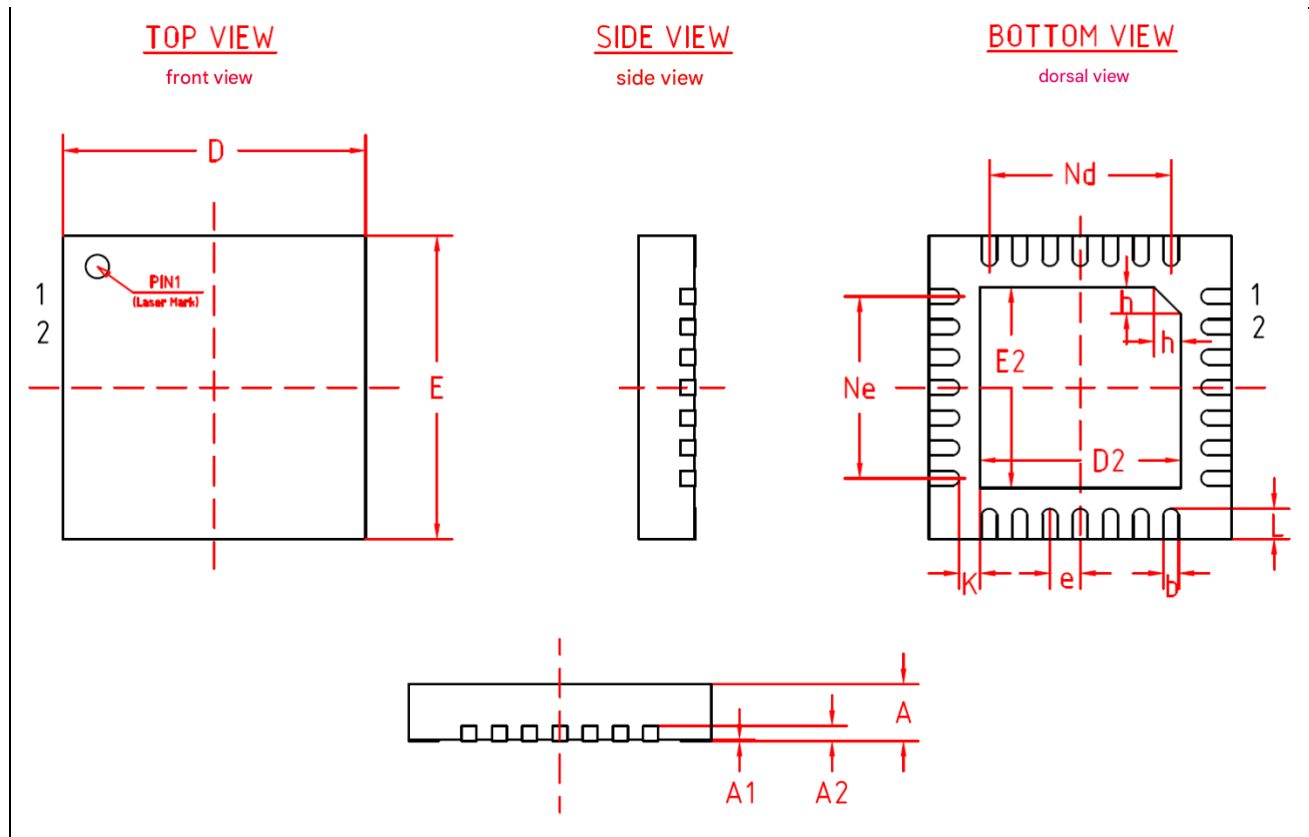


Figure 14-1 QFN28 Package Diagram

Table 14-1 QFN28 package dimensions

Symbol	Minimum value (mm)	Typical value (mm)	Maximum value (mm)
A	0.70	0.75	0.80
A1	-	0.02	0.05
A2	0.203 REF		
b	0.15	0.20	0.25
D	3.90	4.00	4.10
D2	2.55	2.65	2.75
E	3.90	4.00	4.10
E2	2.55	2.65	2.75
e	0.40 BSC		
K	0.225	0.275	0.325
L	0.35	0.40	0.45
h	0.30	0.35	0.40
Ne	2.40 BSC		
Nd	2.40 BSC		

15 Notes

- 1) This product is a CMOS device. Please ensure proper anti-static protection during storage, transportation, and use.
- 2) The device must be properly grounded during use.
- 3) The reflow soldering temperature must not exceed 260°C.

The lead-free reflow soldering process curve is shown below:

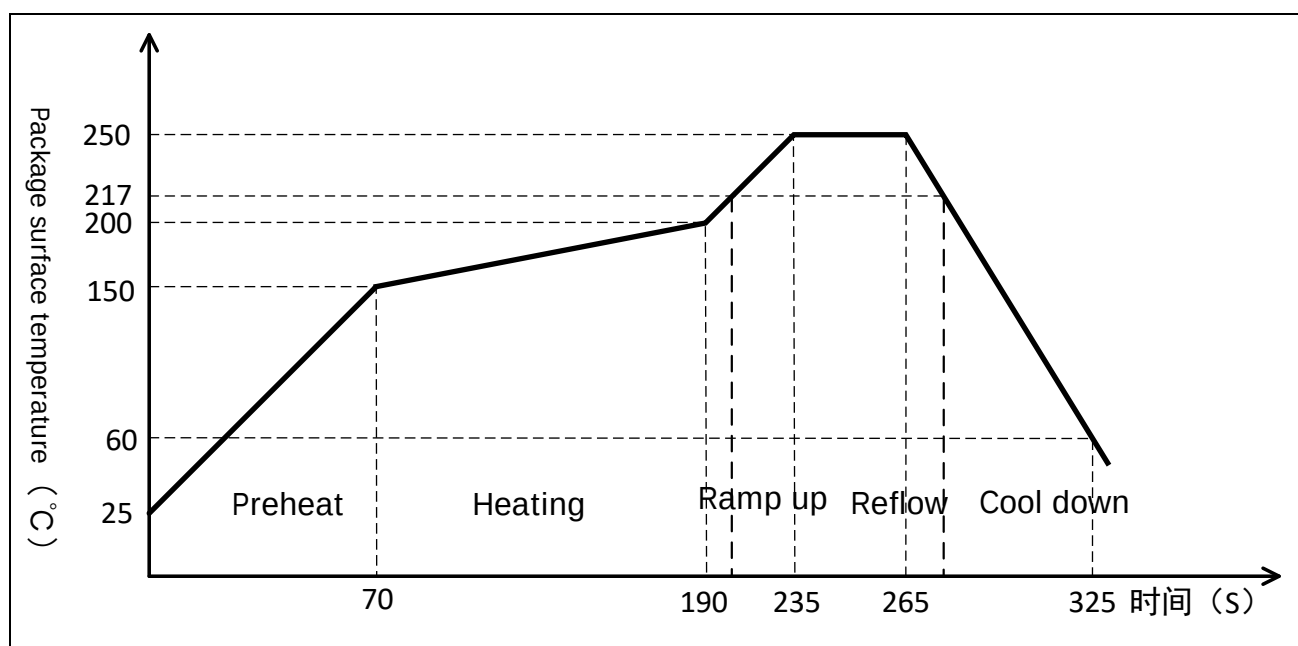


Figure 15-1 Reflow Process Profile

16 Storage conditions

- 1) Products stored in sealed packaging: Up to 12 months at a temperature below 30°C and a humidity below 90%.
- 2) After the packaging is opened, components subjected to reflow soldering or other high-temperature processes must meet the following requirements:
 - a) Keep within 72 hours at a factory temperature below 30°C and $\leq 60\%$ RH;
 - b) Store at 10% RH;
 - c) Bake at 125°C for 24 hours to remove any internal moisture before use.
- 3) MSL (Moisture Sensitivity Level): Level 3 (determined according to IPC/JEDEC J-STD-020).

Abbreviations

ADC	Analog-to-Digital Converter
CAD	Channel Activity Detection
Chirp	Linear Frequency Modulation
CRC	Cyclic Redundancy Check
CSN	SPI Chip Select
DAC	Digital-to-Analog Converter
DCDC	Direct Current Converter
FIFO	First-In First-Out
GPIO	General-Purpose Input/Output
IRQ	Interrupt Request
LDO	Low-Dropout Regulator
LPF	Low-Pass Filter
MAC	Media Access Control
MCU	Microprocessor Unit
Mixer	Mixer
Modem	Modem
OSC	Oscillator
PA	Power Amplifier
RF	Radio Frequency
PLL	Phase-Locked Loop
PMU	Power Management Unit
POR	Power-On Reset
RAM	Random Access Memory
RSSI	Received Signal Strength Indicator
SCK	SPI Clock
SF	Spreading Factor
SPI	Serial Peripheral Interface
STB	Standby Mode
Sync	Synchronization
VCO	Voltage-Controlled Oscillator

Revision History

Version	Date	Content
V1.0	2023.05	First edition
V1.1	2023.06	Added packaging method, reference schematic, optimized pin1 and pin2 names in pin definition, and the function remained unchanged
V1.2	2023.07	Added PAN3060 series, electrical characteristics, etc.
V1.3	2023.08	Updated reference schematic, supplemented working voltage description
V1.4	2023.09	Supplemented reference schematics for different frequency bands, supplemented electrical characteristics for different frequency bands
V1.5	2023.10	Updated reference schematic
V1.6	2023.12	Updated the description of SYNCWORD register in basic operation configuration, updated reference schematic
V1.7	2024.03	Updated reference schematic and electrical characteristics.
V1.8	2024.05	Added description of NC in pin description, updated sleep current, transmission current, updated working frequency band
V1.9	2024.08	Increase the operating frequency band

Document Description

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